

Request by Alexander Braun from 18-Feb-2026

Interrupted Time Series - Segmented linear regression // Nonparametric regression

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22 June 2026

Preliminary/Which data do I use?

Update 22-Jun-2026. Based on the reviewers comments we received on 22 June 2026, i.e.,

- Day-of-week effects: Daily data over four years will exhibit strong within-week periodicity (weekday vs. weekend admission volumes differ substantially). This is not modelled. Day-of-week fixed effects should be included, or the authors must justify their omission.
- Public holidays and the Christmas/New Year period: The interruption point (January 2024) coincides with a known seasonal peak in ED presentations (cited references 29 and 30). The influenza adjustment does not capture post-holiday effects. A holiday dummy should be considered.

I have rerun the ITS regression analysis accounting for holiday and day of the week. To this end, Alexander Braun has sent me an Excel sheet `Feiertage_AT_Taggenau_2022_2025.xlsx`.

$$y_t = \beta_0 + \beta_1 t + \beta_2 I_t + \beta_3 (t - T) I_t + \beta_4 F_t + \beta_5 H_t + \gamma_k W D_{k_t} + \varepsilon_t,$$

where

- y_t is the number of hospital admissions at day $t = 1, \dots, n$;
- T is the interruption day, in our case $T = 731$,
- I_t is an indicator variable which takes 0 before the interruption, i.e., if $t < T$, and 1 otherwise,
- F_t is an indicator variable (whether the day is during influenza period (1=yes, 0=no))
- H_t is an indicator variable (whether the day is a public holiday in Austria (1=yes, 0=no))
- WD_{k_t} are six indicator variables $k = 2, \dots, 7$ (whether the day is a particular day of the week (Tuesday,...,Sunday; yes=1, no=0)).
- ε_t are modelled as autocorrelated errors of order 1.

Update 17-Jun-2026. Based on the reviewer comments from 15 June 2026, we were asked to

- run the regression again excluding weekends as a robustness check
 - use weekend ED admission rates to check the plausibility of their estimated counterfactual trend,
- perform an additional analysis distinguishing between visits during the opening hours of the PHCC,
 - that is, when it was actually available as a substitute.

In this regard, I have performed the ITS and the nonparametric trend analysis again, treating Saturdays and Sundays as missing values, i.e., I have deleted the corresponding rows from the data set. Conversely, I have also deleted all weekdays (Monday - Friday), i.e., only kept the rows with Saturdays and Sundays and performed the ITS and the nonparametric trend analysis on that reduced data set.

As part of the descriptive analysis, I have included average daily visits and yearly visits by day of the week. Moreover, I have included average daily visits by day of the week and whether the the PHCC was open or not (**PVE_open**). In addition, I have also performed a nonparametric trend series estimation for visits during the PPHC opening hours only.

Update 13-Apr-2026. Based on the manuscript from Alexander Braun (received by e-mail on 8-Apr-2026 at 18:17), I have added some additional computations.

Update 03-Apr-2026. Alexander Braun sent me an email at 09:08 with the request to add three robustness checks.

- Eine Variation des Interruptionpoints +/- 1 Monat und einen mit 1.1.2025

- Evtl. einer anderen Autokorrelationsannahme ($Ar(2)$)?
- Einer parametrischen Schätzung mit den Monatswerten? (Vermutlich geht da die Influenzaadjustierung nicht)

Update 24-Mar-2026. Alexander Braun sent me an email at 10:47 with a link to the complete data set, which I downloaded from <https://donauuni.sharepoint.com/:x:/s/EbVF/IQAin0Q910FBTJlyLZaEDy60AXy2zfBWdV3r7PPKEvXKc8U?e=D2RyRy> on 25-Mar-2026 at 09:05. In his email he explained/requested

- In Patient Age sind 16 un plausible Werte (Geburtsdatum nach Aufnahme), ich habe die Werte aus Patient_Age herausgenommen.
- Es gibt einige doppelte Fälle, die komplett identisch sind. Kannst Du die identifizieren und nur einmal zählen?
- PVE_Open ist nun angepasst. Nur WAHRE Fälle sind bezeichnet mit WAHR (dort wo es das Zentrum nicht gab oder keine Öffnungszeit war, ist die Zelle leer)
- Für die ITSA wären drei Tabellen vorgesehen + die Grafiken (einmal parametric und einmal non-parametric und evtl. in der X-Achse die Jahreszahlen anstatt die laufenden Tage).

Update 11-Mar-2026. Veronika Boskova gave me an article by

- Kovacevic et al. (2025). **The impact of primary care networks on emergency hospitalisations in the English NHS: An interrupted time series analysis.** Health policy. <https://doi.org/10.1016/j.healthpol.2025.105524>.

I have some doubts as to whether this approach is preferable because of overfitting issues. It is my understanding that they have essentially 26 data points (quarters) in total and model two interruptions and seasonality as binary dummy variables on top.

Update 04-Mar-2026. Further to today's meeting with Alexander Braun, I incorporate the possibility of a different intercept during influenza season in the ITS equation.

17-Feb-2026. I received an email from Alexander Braun with a request to perform an Interrupted Time Series Analysis. I used the data as downloaded from MS Teams <https://donauuni.sharepoint.com/:x:/s/EbVF/IQCYSk6ZYf8eQoPEiYvG5FNEAc0cXjS0hHeHlnqnSbAXgM4?e=pjcRSb> on 18-Feb-2026 at 08:52 (CET).

This was part of his email:

- Uns geht es darum herauszufinden, ob das neugegründete Primärversorgungszentrum (Öffnung am 1.1.2024) Auswirkungen auf die Notfallbehandlungen im Spital Wiener Neustadt hatten (Jede Zeile ist eine gesonderte Aufnahme).
- Unsere Überlegung dazu war eine ITSA zu berechnen. Wir denken, dass folgende Prädiktoren Auswirkung haben könnten:
 - Ist das PVZ offen (also vorher-nachher und wenn es eröffnet hat, ob Wochenende/Öffnungszeiten ist („Spalte H: PVE_Open“))
 - * Based on previous emails the interruption in the time series (PVE-Eröffnung occurred on 01-Jan-2024).
 - das Alter (Spalte I Patient_Birth_year)
 - das Geschlecht („Spalte L (M/W“)
 - Schweregrad entlang des Manchester-Triage-Systems (vermutlich eher Prio 3 und 4, da diese vermutlich auch in PVE behandelt werden können (z.B. Schnittwunden,...))
 - Ob gerade Grippesaison ist oder nicht (Spalte N: Influenza)
- Ich würde weiters die These aufstellen, dass die Tage von Weihnachten bis Neujahr ebenfalls einen Einfluss hat (Christmas Shock oder wie auch immer man das nennen möchte, zumindest gibt es dazu Literatur).
- Kannst Du damit was anfangen? Ich glaube zu komplex ist das gar nicht, aber ich habe dir ja gezeigt, dass diese Zeit um die Eröffnung halt sehr fuzzy ist, das wir mit Influenza allein nicht erklären konnten. Vielleicht hast Du dazu eine Idee? Ich würde mich freuen, wenn Du mir dabei hilfst.

General Comments from Claus

Interrupted Time Series. For the interrupted time series analysis I used the model by Korevaar et al. (see <https://tinyurl.com/its-meta-analysis-workshop>).

$$y_t = \beta_0 + \beta_1 t + \beta_2 I_t + \beta_3 (t - T) I_t + \varepsilon_t,$$

where

- y_t is the number of hospital admissions at day $t = 1, \dots, n$;
- T is the interruption day, in our case $T = 731$,
- I_t is an indicator variable which takes 0 before the interruption, i.e., if $t < T$, and 1 otherwise,
- ε_t are modelled as autocorrelated errors of order 1.

Note that this approach might not be ideal because y_t are count data so arguably a GLMM with a negative binomial distribution might be more appropriate. However, treating y_t as continuous simplifies interpretation and fluctuations (at least when considering all data) do not seem to substantially deviate from normality.

If we already consider Influenza, I would expect this covariate to be correlated with “Christmas” season, so I am not really sure whether it makes sense to include both in a model.

The outliers seem less pronounced after removing duplicate rows.

Update 04-Mar-2026:

I include an indicator variable F_t which takes 1 if the day t is during the influenza season and 0 otherwise. The coefficient β_4 allows for a different intercept during the influenza season,

$$y_t = \beta_0 + \beta_1 t + \beta_2 I_t + \beta_3 (t - T) I_t + \beta_4 F_t + \varepsilon_t,$$

with the other variables remaining as before. A graphical inspection of the influenza admissions per day seems to support this approach.

Update 31-Mar-2026: As a sensitivity analysis, I also modelled the error terms, i.e., ε_t as an ARMA(1,1) process for the full data accounting for influenza.

Update 03-Apr-2026: I also modelled the errors as AR(2) process as requested by Alexander Braun.

Nonparametric trend estimation. In addition to the Interrupted Time Series analysis, I would recommend a semiparametric model defined by

$$y_t = m(x_t) + \varepsilon_t,$$

where

- y_t is the number of hospital admissions at time point (or day) $t = 1, \dots, n$;
- $x_t = t/n$ denotes the rescaled time,
- m is a smooth function representing the trend, which is estimated in a nonparametric manner,
- ε_t is a zero mean stationary process.

This method is implemented in the R package `smoots` which allows for a completely data driven selection of the optimal bandwidth while taking autocorrelation into account. See also

- Feng Y, Gries T, Letmathe S, Schulz D. The `smoots` Package in R for Semiparametric Modeling of Trend Stationary Time Series. *R Journal* March 2022;14/1:182-195. <https://doi.org/10.32614/rj-2022-017>.

Which packages and R version do I use?

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2     4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr       1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readxl)
library(smoots)
```

```
*****
Welcome to the package 'smoots'!
*****
```

Please report any possible errors and bugs to dominik.schulz@uni-paderborn.de.
Thank you.

```
*****
```

```
library(nlme)
```

Attaching package: 'nlme'

The following object is masked from 'package:dplyr':

collapse

```
library(multcomp)
```

Loading required package: mvtnorm
Loading required package: survival
Loading required package: TH.data
Loading required package: MASS

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

select

Attaching package: 'TH.data'

The following object is masked from 'package:MASS':

geyser

sessionInfo()

R version 4.5.3 (2026-03-11 ucrt)

Platform: x86_64-w64-mingw32/x64

Running under: Windows 10 x64 (build 19045)

Matrix products: default

LAPACK version 3.12.1

locale:

[1] LC_COLLATE=English_Europe.utf8 LC_CTYPE=English_Europe.utf8

[3] LC_MONETARY=English_Europe.utf8 LC_NUMERIC=C

[5] LC_TIME=English_Europe.utf8

time zone: Europe/Vienna

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] multcomp_1.4-30 TH.data_1.1-5 MASS_7.3-65 survival_3.8-6

[5] mvtnorm_1.3-6 nlme_3.1-168 smoots_1.1.4 readxl_1.4.5


```
[9] lubridate_1.9.5 forcats_1.0.1 stringr_1.6.0 dplyr_1.2.0
[13] purrr_1.2.1 readr_2.2.0 tidyr_1.3.2 tibble_3.3.1
[17] ggplot2_4.0.2 tidyverse_2.0.0
```

loaded via a namespace (and not attached):

```
[1] gtable_0.3.6 xfun_0.56 lattice_0.22-9
[4] tzdb_0.5.0 vctrs_0.7.1 tools_4.5.3
[7] generics_0.1.4 parallel_4.5.3 sandwich_3.1-1
[10] pkgconfig_2.0.3 Matrix_1.7-4 RColorBrewer_1.1-3
[13] S7_0.2.1 lifecycle_1.0.5 compiler_4.5.3
[16] farver_2.1.2 progress_1.2.3 codetools_0.2-20
[19] htmltools_0.5.9 yaml_2.3.12 pillar_1.11.1
[22] crayon_1.5.3 parallelly_1.46.1 tidyselect_1.2.1
[25] digest_0.6.39 stringi_1.8.7 future_1.70.0
[28] listenv_0.10.1 splines_4.5.3 fastmap_1.2.0
[31] grid_4.5.3 cli_3.6.5 magrittr_2.0.4
[34] future.apply_1.20.2 withr_3.0.3 prettyunits_1.2.0
[37] scales_1.4.0 timechange_0.4.0 rmarkdown_2.30
[40] globals_0.19.1 otel_0.2.0 progressr_0.18.0
[43] cellranger_1.1.0 zoo_1.8-15 hms_1.1.4
[46] evaluate_1.0.5 knitr_1.51 rlang_1.1.7
[49] Rcpp_1.1.1-1.1 glue_1.8.0 rstudioapi_0.19.0
[52] jsonlite_2.0.0 R6_2.6.1
```

How do read in the data set and get the dates for the Influenza seasons?

```
filename <- "C:/Users/CNowak/OneDrive - Universität für Weiterbildung Krems/Beratung/2026/AlexanderBraun_20260217/"

# all_dates <- tibble(Triage_Begin_Date = seq.Date(as.Date("2022-01-01"),
#                                                  as.Date("2025-12-31"),
```

```

#                                     by = "day"))

addVariables <- function(dat,intday="01Jan2024"){
  dat <- alldates |> left_join(dat,by=c("Triage_Begin_Date",
                                     "Day",
                                     "Month",
                                     "Year",
                                     "Week_Day"))

  dat <- dat |> mutate(
    Triage_Begin_Day=as.integer(Triage_Begin_Date - min(Triage_Begin_Date)) + 1,
    Interruption_Day=as.integer(as.Date(x=intday,"%d%b%Y") - min(Triage_Begin_Date)) + 1,
    Level_Change=ifelse(Triage_Begin_Day < Interruption_Day,0,1),
    Slope_Change=ifelse(Level_Change==0,0,Triage_Begin_Day-Interruption_Day), #01Jan2024 equals Triage_Beg
    n=ifelse(is.na(n),0,n),
    .after=1)

  return(dat)
}

addVariables2 <- function(dat,intday="01Jan2024"){
  dat <- dat |> mutate(
    Triage_Begin_Month=as.integer(interval(min(Triage_Begin_Date),Triage_Begin_Date) %/% months(1)) + 1,
    Interruption_Month=as.integer(interval(min(Triage_Begin_Date),as.Date(x=intday,"%d%b%Y")) %/% months(1)
    Level_Change=ifelse(Triage_Begin_Month < Interruption_Month,0,1),
    Slope_Change=ifelse(Level_Change==0,0,Triage_Begin_Month-Interruption_Month),
    n=ifelse(is.na(n),0,n),
    .after=1)

  return(dat)
}

weekdaylevels <- c("Mon","Tue","Wed","Thu","Fri","Sat","Sun")

```

```

holidays <- read_xlsx(str_c(filename,"Feiertage_AT_Taggenau_2022_2025.xlsx")) |>
  mutate(Triage_Begin_Date=as_date(dmy(Datum)),
         Holiday=ifelse(Feiertag==TRUE,1,0)) |>
  dplyr::select(Triage_Begin_Date,Holiday)

dat00 <- read_xlsx(str_c(filename,"MTS Daten_2022-2025_complete.xlsx"),
                  range="A1:M84739",
                  sheet="Daten2022-2025") |>
  mutate(Triage_Begin_Date=as_date(Triage_Begin), .after=1) |>
  distinct() |>
  mutate(WD = recode(Week_Day,"Mo" = "Mon",
                      "Di" = "Tue",
                      "Mi" = "Wed",
                      "Do" = "Thu",
                      "Fr" = "Fri",
                      "Sa" = "Sat",
                      "So" = "Sun"),
         WD = fct_relevel(WD,weekdaylevels),
         WD2 = ifelse(WD=="Tue",1,0),
         WD3 = ifelse(WD=="Wed",1,0),
         WD4 = ifelse(WD=="Thu",1,0),
         WD5 = ifelse(WD=="Fri",1,0),
         WD6 = ifelse(WD=="Sat",1,0),
         WD7 = ifelse(WD=="Sun",1,0)) |>
  left_join(holidays,by="Triage_Begin_Date")

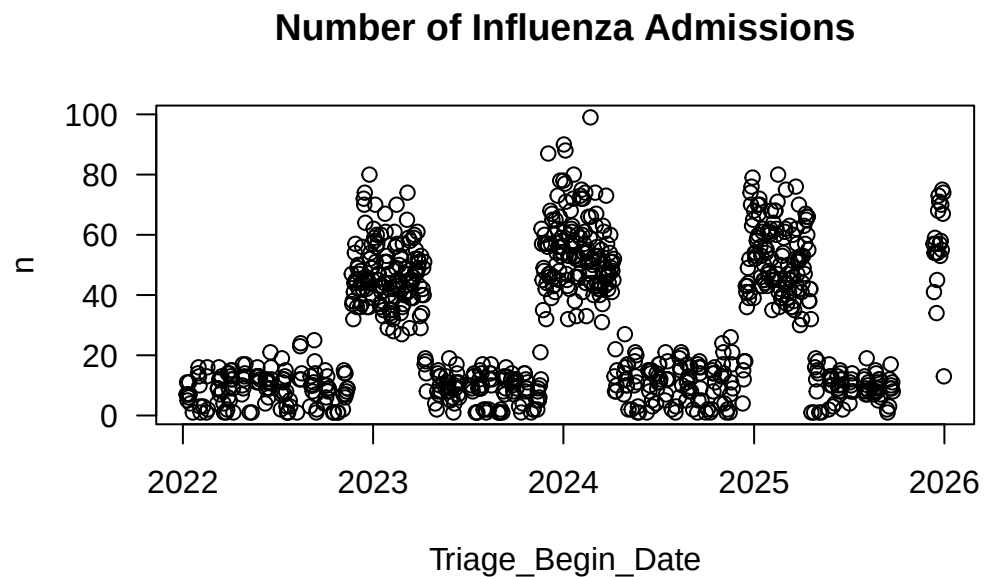
alldates <- dat00 |> group_by(Triage_Begin_Date,Day,Month,Year,
                             Week_Day) |>
  count() |> ungroup() |>

```

```
dplyr::select(-n)

influenz <- dat00 |> dplyr::select(Triage_Begin_Date,Influenza) |>
  filter(Influenza=="ja") |> count(Triage_Begin_Date)

plot(influenz,las=1,main="Number of Influenza Admissions")
```



```
influenz |> filter(n>=30) |>
  mutate(next_date = lead(Triage_Begin_Date),
```

```
gap = as.integer(next_date - Triage_Begin_Date)) |>
filter(gap >= 2)
```

A tibble: 8 x 4

	Triage_Begin_Date	n	next_date	gap
	<date>	<int>	<date>	<int>
1	2023-01-28	51	2023-01-30	2
2	2023-02-08	32	2023-02-10	2
3	2023-02-24	52	2023-02-26	2
4	2023-03-11	45	2023-03-13	2
5	2023-04-01	33	2023-04-03	2
6	2023-04-09	51	2023-11-20	225
7	2024-04-07	52	2024-12-16	253
8	2025-04-20	32	2025-12-11	235

```
influenz |> filter(n>=30) |>
summarise(season1_start=min(Triage_Begin_Date),
season4_end=max(Triage_Begin_Date))
```

A tibble: 1 x 2

	season1_start	season4_end
	<date>	<date>
1	2022-11-21	2025-12-30

```
alldates <- alldates |>
mutate(season1=as.integer(Triage_Begin_Date >= as.Date("2022-11-21") & Triage_Begin_Date <= as.Date("2023-04-09")),
season2=as.integer(Triage_Begin_Date >= as.Date("2023-11-20") & Triage_Begin_Date <= as.Date("2024-04-07")),
season3=as.integer(Triage_Begin_Date >= as.Date("2024-12-16") & Triage_Begin_Date <= as.Date("2025-04-20")),
season4=as.integer(Triage_Begin_Date >= as.Date("2025-12-11") & Triage_Begin_Date <= as.Date("2025-12-31")),
```

```
influenza=season1+season2+season3+season4) |>
dplyr::select(-c(season1,season2,season3,season4))
```

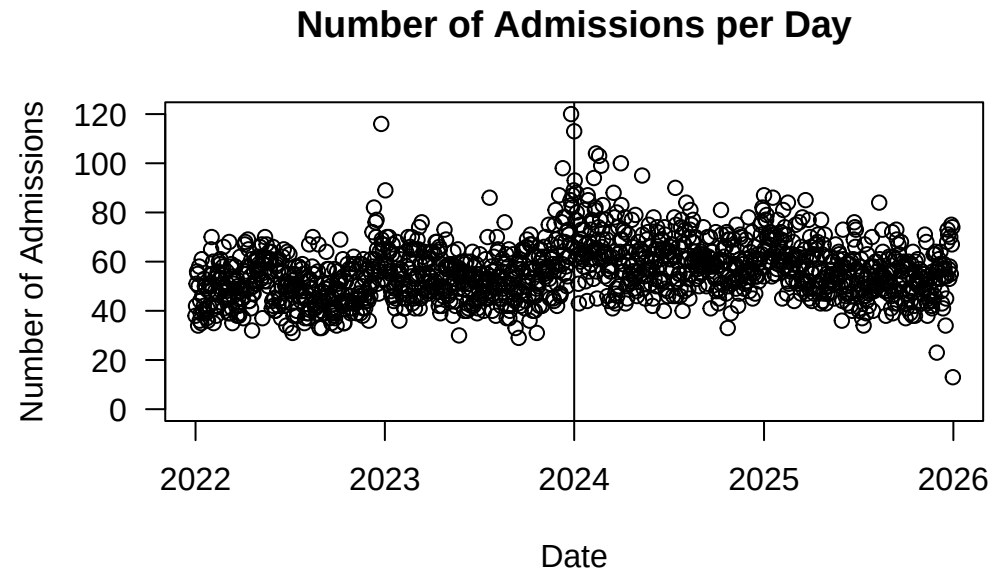
Based on this data set I extracted the following dates for the influenza season:

Influenza	Start Date	End Date
Season 1	2022-11-21	2023-04-09
Season 2	2023-11-20	2024-04-07
Season 3	2024-12-16	2025-04-20
Season 4	2025-12-11	2025-12-31

How do I get daily and monthly admissions before and after the opening of the PHC?

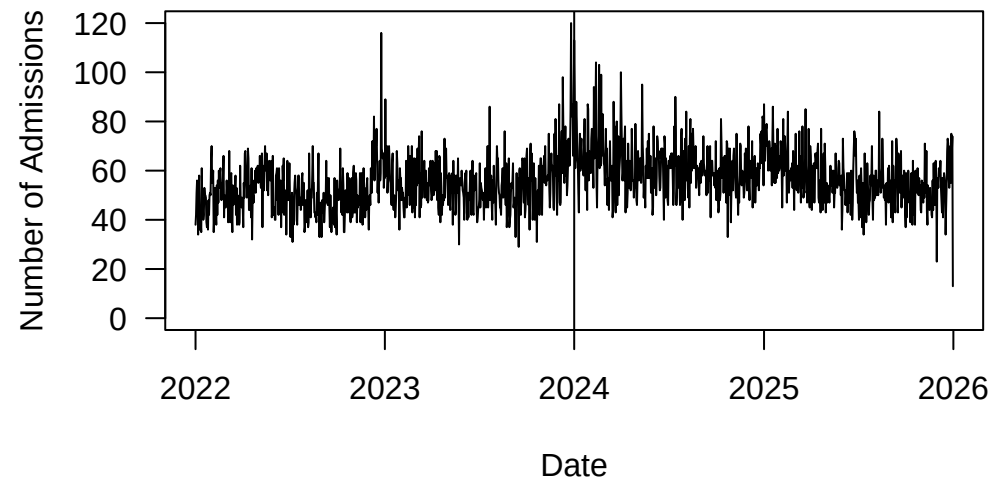
```
daily <- dat00 |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day,WD) |>
  count() |>
  mutate(PreInterruption = ifelse(Triage_Begin_Date < as.Date(x="01Jan2024", "%d%b%Y"),0,1))

plot(daily$Triage_Begin_Date,daily$n,
      ylim=c(0,120),
      las=1,
      ylab="Number of Admissions",
      #type="l",
      xlab="Date",
      main="Number of Admissions per Day")
abline(v=as.Date(x="01Jan2024", "%d%b%Y"))
```



```
plot(daily$Triage_Begin_Date,daily$n,  
      ylim=c(0,120),  
      las=1,  
      ylab="Number of Admissions",  
      type="l",  
      xlab="Date",  
      main="Number of Admissions per Day")  
abline(v=as.Date(x="01Jan2024", "%d%b%Y"))
```

Number of Admissions per Day



```
monthly <- dat00 |>
  mutate(Triage_Begin_ym=format(Triage_Begin, "%Y-%m"),
         Triage_Begin_ymf=floor_date(Triage_Begin_Date,unit="month"),.after=2) |>
  group_by(Year,Triage_Begin_ymf) |>
  count() |>
  mutate(PreInterruption = ifelse(Triage_Begin_ymf < as.Date(x="01Jan2024", "%d%b%Y"),0,1))

# monthly2<- dat00 |>
#   mutate(Triage_Begin_ym=format(Triage_Begin, "%Y-%m"),
#          Triage_Begin_ymf=floor_date(Triage_Begin_Date,unit="month"),.after=2) |>
#   group_by(Year,Triage_Begin_ymf,WD) |>
#   count() |>
```



```

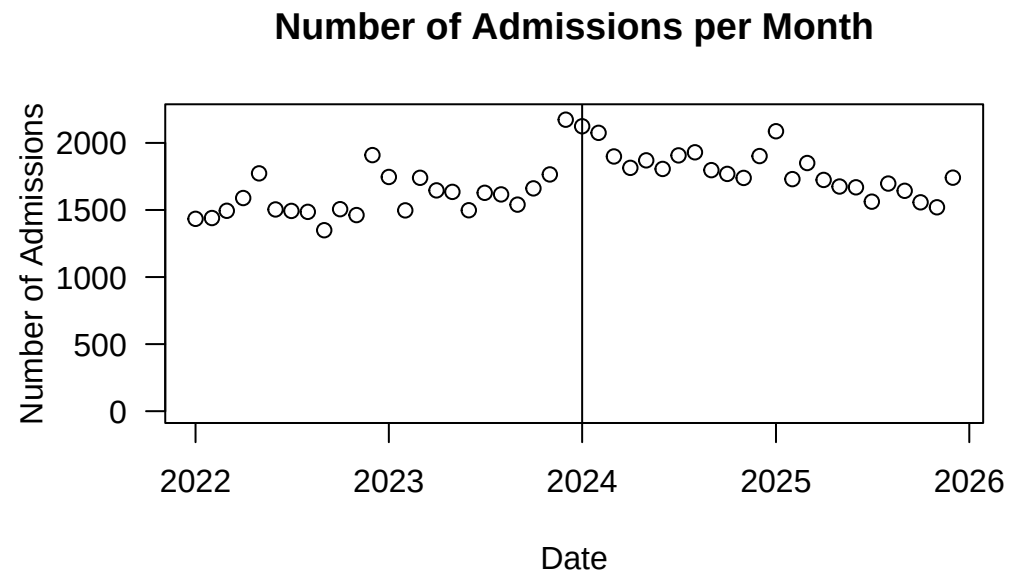
#           mutate(PreInterruption = ifelse(Triage_Begin_ymf < as.Date(x="01Jan2024", "%d%b%Y"), 0, 1))

yearly <- dat00 |>
  group_by(Year) |>
  count()

yearly2 <- dat00 |>
  group_by(Year, WD) |>
  count()

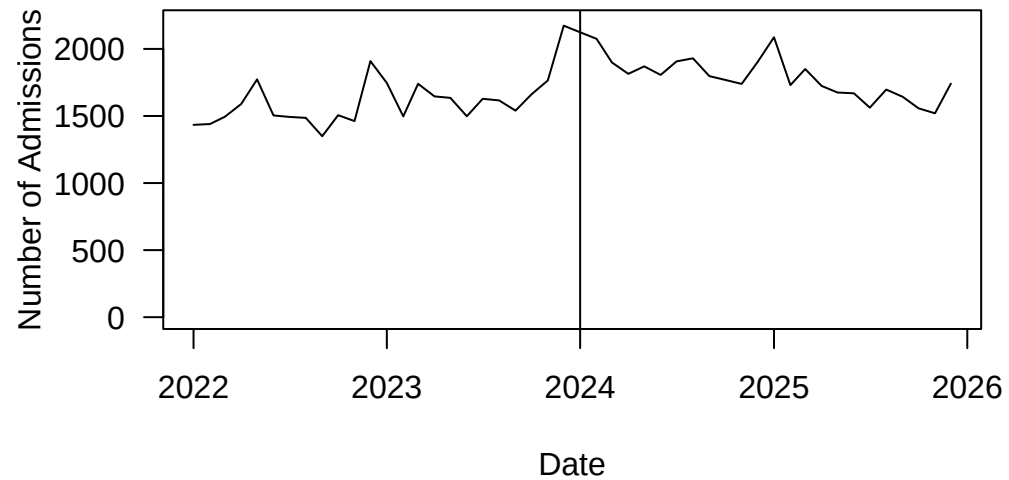
plot(monthly$Triage_Begin_ymf, monthly$n,
      ylim=c(0, 2200),
      las=1,
      ylab="Number of Admissions",
      #type="l",
      xlab="Date",
      main="Number of Admissions per Month")
abline(v=as.Date(x="01Jan2024", "%d%b%Y"))

```



```
plot(monthly$Triage_Begin_ymf,monthly$n,  
      ylim=c(0,2200),  
      las=1,  
      ylab="Number of Admissions",  
      type="l",  
      xlab="Date",  
      main="Number of Admissions per Month")  
abline(v=as.Date(x="01Jan2024", "%d%b%Y"))
```

Number of Admissions per Month



```
desdaily <- daily |> group_by(PreInterruption) |>
  summarise(mean=mean(n),
            sd=sd(n),
            n=n())

desdaily2 <- daily |> group_by(PreInterruption,WD) |>
  summarise(mean=mean(n),
            sd=sd(n),
            n=n())
```

`summarise()` has regrouped the output.

i Summaries were computed grouped by PreInterruption and WD.
 i Output is grouped by PreInterruption.
 i Use `summarise(.groups = "drop\_last")` to silence this message.
 i Use `summarise(.by = c(PreInterruption, WD))` for per-operation grouping
 (`?dplyr::dplyr\_by`) instead.

```
desdaily3 <- dat00 |> group_by(Triage_Begin_Date,Day,Month,Year,Week_Day,WD,PVE_Open) |>
  count() |>
  mutate(PreInterruption = ifelse(Triage_Begin_Date < as.Date(x="01Jan2024", "%d%b%Y"),0,1)) |>
  group_by(PreInterruption,WD,PVE_Open) |>
  summarise(mean=mean(n),
            sd=sd(n),
            n=n())
```

`summarise()` has regrouped the output.
 i Summaries were computed grouped by PreInterruption, WD, and PVE_Open.
 i Output is grouped by PreInterruption and WD.
 i Use `summarise(.groups = "drop\_last")` to silence this message.
 i Use `summarise(.by = c(PreInterruption, WD, PVE\_Open))` for per-operation
 grouping (`?dplyr::dplyr\_by`) instead.

```
desmonthly <- monthly |> group_by(PreInterruption) |>
  summarise(mean=mean(n),
            sd=sd(n),
            n=n())
```

```
desdaily
```

```
# A tibble: 2 x 4
```

	PreInterruption	mean	sd	n
	<dbl>	<dbl>	<dbl>	<int>
1	0	52.9	10.7	730
2	1	58.9	11.4	731

```
desmonthly
```

```
# A tibble: 2 x 4
```

	PreInterruption	mean	sd	n
	<dbl>	<dbl>	<dbl>	<int>
1	0	1608.	180.	24
2	1	1795.	161.	24

```
yearly
```

```
# A tibble: 4 x 2
```

```
# Groups:   Year [4]
```

	Year	n
	<dbl>	<int>
1	2022	18439
2	2023	20145
3	2024	22632
4	2025	20455

```
(desdaily$mean[2]-desdaily$mean[1])/desdaily$mean[1]*100 # change in percent
```

```
[1] 11.51788
```

```
(desmonthly$mean[2]-desmonthly$mean[1])/desmonthly$mean[1]*100 # change in percent
```

```
[1] 11.67064
```

```
desdaily2
```

```
# A tibble: 14 x 5
```

```
# Groups:   PreInterruption [2]
```

	PreInterruption	WD	mean	sd	n
	<dbl>	<fct>	<dbl>	<dbl>	<int>
1	0	Mon	57.7	9.36	104
2	0	Tue	52.3	10.9	104
3	0	Wed	52.1	9.60	104
4	0	Thu	51.5	9.88	104
5	0	Fri	52.9	9.47	104
6	0	Sat	50.2	10.1	105
7	0	Sun	53.3	13.8	105
8	1	Mon	65.1	10.2	105
9	1	Tue	55.9	9.38	105
10	1	Wed	55.3	10.5	105
11	1	Thu	57.1	9.83	104
12	1	Fri	58.0	10.8	104
13	1	Sat	59.6	11.7	104
14	1	Sun	61.6	14.0	104

```
yearly2
```

```
# A tibble: 28 x 3
```

```
# Groups:   Year, WD [28]
```

```

      Year WD      n
  <dbl> <fct> <int>
1  2022 Mon     2850
2  2022 Tue     2623
3  2022 Wed     2637
4  2022 Thu     2634
5  2022 Fri     2649
6  2022 Sat     2512
7  2022 Sun     2534
8  2023 Mon     3147
9  2023 Tue     2818
10 2023 Wed     2781
# i 18 more rows

```

```
desdaily3
```

```

# A tibble: 19 x 6
# Groups:   PreInterruption, WD [14]
  PreInterruption WD PVE_Open mean sd n
      <dbl> <fct> <lgl>    <dbl> <dbl> <int>
1           0 Mon  NA      57.7  9.36  104
2           0 Tue  NA      52.3 10.9   104
3           0 Wed  NA      52.1  9.60  104
4           0 Thu  NA      51.5  9.88  104
5           0 Fri  NA      52.9  9.47  104
6           0 Sat  NA      50.2 10.1   105
7           0 Sun  NA      53.3 13.8   105
8           1 Mon  TRUE     46.8  8.08  105
9           1 Mon  NA      18.3  4.92  105
10          1 Tue  TRUE     39.3  7.76  105
11          1 Tue  NA      16.6  4.40  105

```

12	1	Wed	TRUE	38.6	8.25	105
13	1	Wed	NA	16.7	4.96	105
14	1	Thu	TRUE	40.5	8.12	104
15	1	Thu	NA	16.6	4.54	104
16	1	Fri	TRUE	40.2	8.51	104
17	1	Fri	NA	17.8	5.27	104
18	1	Sat	NA	59.6	11.7	104
19	1	Sun	NA	61.6	14.0	104

How do I get additional results for the manuscript from 08-Apr-2026?

```
dat00 |> dplyr::select(Year, Patient_Age) |>
  group_by(Year) |>
  summarize(mean=mean(Patient_Age, na.rm=T),
            sd=sd(Patient_Age, na.rm=T))
```

```
# A tibble: 4 x 3
  Year mean  sd
<dbl> <dbl> <dbl>
1  2022  56.5  25.2
2  2023  57.5  23.2
3  2024  58.8  20.7
4  2025  61.5  19.1
```

```
dat00 |> group_by(Year) |>
  count(Prio) |>
  mutate(Percent=round(n/sum(n)*100,2))
```



```
# A tibble: 20 x 4
# Groups:   Year [4]
   Year Prio     n Percent
  <dbl> <dbl> <int>   <dbl>
1  2022     1     66    0.36
2  2022     2    2086   11.3
3  2022     3    6715   36.4
4  2022     4    8396   45.5
5  2022     5    1176    6.38
6  2023     1     49    0.24
7  2023     2    2120   10.5
8  2023     3    8079   40.1
9  2023     4    8920   44.3
10 2023     5     977    4.85
11 2024     1     46    0.2
12 2024     2    2315   10.2
13 2024     3    8967   39.6
14 2024     4    9834   43.4
15 2024     5    1470    6.5
16 2025     1     43    0.21
17 2025     2    1985    9.7
18 2025     3    8231   40.2
19 2025     4    8988   43.9
20 2025     5    1208    5.91
```

How do I get the data sets for the analyses?

```
datall<- dat00 |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
```

```

        count() |> ungroup() |> addVariables()

datalld<-dat00 |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables(intday="01Dec2023")

datallf<-dat00 |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables(intday="01Feb2024")

datall5<-dat00 |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables(intday="01Jan2025")

datallm<-dat00 |>
  mutate(Triage_Begin_Date = floor_date(Triage_Begin_Date,"month")) |>
  dplyr::select(Triage_Begin_Date,Month,Year) |>
  group_by(Triage_Begin_Date,Month,Year) |>
  count() |> ungroup() |> addVariables2(intday="01Jan2024")

##### Subgroup analyses #####
datsem<- dat00 |> filter(Patient_Sex=="M") |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables()

datsef<- dat00 |> filter(Patient_Sex=="W") |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables()

datag12<-dat00 |> filter(Patient_Age_Category %in% c("1","2")) |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>

```

```

        count() |> ungroup() |> addVariables()

datag3<- dat00 |> filter(Patient_Age_Category=="3") |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables()

datag4<- dat00 |> filter(Patient_Age_Category=="4") |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables()

datag5<- dat00 |> filter(Patient_Age_Category=="5") |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables()

datpr12<-dat00 |> filter(Prio %in% c("1","2")) |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables()

datpr3<- dat00 |> filter(Prio=="3") |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables()

datpr45<-dat00 |> filter(Prio %in% c("4","5")) |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day) |>
  count() |> ungroup() |> addVariables()

##### Add subgroup analyses for reviewer #####
datwk <-dat00 |> filter(WD %in% c("Mon","Tue","Wed","Thu","Fri")) |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day,WD) |>
  count() |> ungroup() |> addVariables() |>
  filter(!is.na(WD))

```

```

datss <-dat00 |> filter(WD %in% c("Sat","Sun")) |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day,WD) |>
  count() |> ungroup() |> addVariables() |>
  filter(!is.na(WD))

datop <-dat00 |> filter(PVE_Open==TRUE) |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day,WD) |>
  count() |> ungroup() |> addVariables() |>
  filter(!is.na(WD))

datopn <-dat00 |> filter(is.na(PVE_Open)) |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day,WD) |>
  count() |> ungroup() |> addVariables() |>
  filter(Level_Change==1)

##### Add analyses for reviewer comments from 22-Jun-2026 #####
datnew <- dat00 |>
  group_by(Triage_Begin_Date,Day,Month,Year,Week_Day,Holiday,WD2,WD3,WD4,WD5,WD6,WD7) |>
  count() |> ungroup() |> addVariables()

```

Which function do I write to perform the segmented regression analysis as suggested by Korevaar et al.?

```

getITSresults <- function(dat0,outcome,ptitle,intday="01Jan2024"){
  form <- formula(paste0(outcome, " ~ Triage_Begin_Day + Level_Change + Slope_Change",sep=""))
  print(form)
  REML_model <- gls(model=form,
    method="REML",

```

```

        corr=corAR1(form=~Triage_Begin_Day),
        data=dat0
    )
    # print the results of the analysis to the console
    print(summary(REML_model)) # display the parameter estimates

    # to save the parameter estimates:
    # predicted values
    model_estimates <- REML_model$fitted
    # save the parameter estimates for display in graphs etc.
    model_intercept <- REML_model$coefficients[1]
    model_pre_slope <- REML_model$coefficients[2]
    model_level_change <- REML_model$coefficients[3]
    model_slope_change <- REML_model$coefficients[4]
    # generate the confidence intervals
    model_CIs <- confint(REML_model)
    # display the confidence intervals in the console
    print("Below are the model estimates together with their 95% confidence limits.")
    print(cbind(`Point estimates`=c(model_intercept,
                                    model_pre_slope,
                                    model_level_change,
                                    model_slope_change),

                model_CIs))
    # save the CIs for display in graphs etc.
    #####
    print("Below are the model estimates together with multiplicity adjusted p-values and confidence limits")
    set.seed(12345)
    REML_modelg <- glht(REML_model)
    print(summary(REML_modelg))
    print(confint(REML_modelg))
    ## outputs Alex:

```

```

model_n    <- REML_model$dims$N
ss         <- summary(REML_model)
model_AIC  <- ss$AIC
model_BIC  <- ss$BIC
print("Below is the output requested by Alexander Braun")
print(round(t(data.frame(model_intercept=model_intercept,
                        model_pre_slope=model_pre_slope,
                        model_level_change=model_level_change,
                        model_slope_change=model_slope_change,
                        model_AIC=model_AIC,
                        model_BIC=model_BIC,
                        model_n=model_n)),5))

#####
### Create a plot ###
#####
dat0$prediction <- predict(REML_model)

# create the counterfactual data, recall this is what would have happened had there been no interruption
# and therefore is simple to generate by just using the intercept and pre_slope data
dat0$counterfactual <- model_intercept + model_pre_slope*dat0$Triage_Begin_Day

# for the graph we need the interruption time
int_time <- as.Date(x=intday,"%d%b%Y")

# colourblind colour blue
myblue <- rgb(0,114,178,max=255,alpha=255)
myothe <- rgb(248,118,109,max=255,alpha=255)

# legends can be tricky, I found this useful... https://stackoverflow.com/questions/17148679/construct-a-manual-legend
# set the colours to be used
colors <- c("Trend lines"=myblue,"Counterfactual"=myblue,"Data points"=myblue)

```

```

# set the line types (the lty values)
line_types <- c("Trend lines"=1,"Counterfactual"=2)

# set up the plot
ITS_plot <- ggplot(data=dat0,aes(x=Triage_Begin_Date,y=!!sym(outcome))) +
  # add the actual data points
  geom_point(color="black", pch=1, size=1, stroke=1) +
  # set the predicted line in the pre-interruption segment
  geom_line(data=subset(dat0,Level_Change==0), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Trend lines")) +
  # set the predicted line in the post-interruption segment
  geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Trend lines")) +
  # set the counterfactual line in the post-interruption segment
  geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=counterfactual,color="Counterfactual",linetype="Counterfactual")) +
  # add some labels to the axes
  labs(x="Date",y="Number of Admissions per Day",title=ptitle) +
  scale_y_continuous(limits = c(0, 120)) +
  # set the colours to be used using the vector we set up above
  scale_color_manual(name="Legend:",values=c(myothe,myblue)) +
  # set up the line types using the vector we set up above
  scale_linetype_manual(name="Legend:",values=line_types) +
  # add a vertical line for the interruption time
  geom_vline(xintercept=int_time, color="black", lty="dashed", lwd=2) +
  theme(legend.position = "bottom",
        #legend.background = element_rect(fill="white",linewidth=1,linetype="solid"),
        #legend.title = element_text(size=15),
        #legend.text = element_text(size=15),
        #legend.key.width = unit(4,"cm"),
        #legend.key = element_rect(colour=NA, fill=NA),
        #panel.background = element_blank(),
        #panel.grid.major = element_line(colour="gray"),
        #axis.title.y = element_text(face="bold", size=20),

```

```

        #axis.title.x = element_text(face="bold", size=20),
        #plot.title = element_text(face="bold", size=20),
        #axis.text.x=element_text(face="bold", size=15),
        #axis.text.y=element_text(face="bold", size=15),
    )
    # show this plot in the plots window
    ITS_plot
}
###
###
###
getITSresultsMonth <- function(dat0,outcome,ptitle,intday="01Jan2024"){
  form <- formula(paste0(outcome, " ~ Triage_Begin_Month + Level_Change + Slope_Change",sep=""))
  print(form)
  REML_model <- gls(model=form,
                    method="REML",
                    corr=corAR1(form=~Triage_Begin_Month),
                    data=dat0
  )
  # print the results of the analysis to the console
  print(summary(REML_model)) # display the parameter estimates

  # to save the parameter estimates:
  # predicted values
  model_estimates <- REML_model$fitted
  # save the parameter estimates for display in graphs etc.
  model_intercept <- REML_model$coefficients[1]
  model_pre_slope <- REML_model$coefficients[2]
  model_level_change <- REML_model$coefficients[3]
  model_slope_change <- REML_model$coefficients[4]
  # generate the confidence intervals

```



```

model_CIs <- confint(REML_model)
# display the confidence intervals in the console
print("Below are the model estimates together with their 95% confidence limits.")
print(cbind(`Point estimates`=c(model_intercept,
                                model_pre_slope,
                                model_level_change,
                                model_slope_change),

            model_CIs))
# save the CIs for display in graphs etc.
####
print("Below are the model estimates together with multiplicity adjusted p-values and confidence limits")
set.seed(12345)
REML_modelg <- glht(REML_model)
print(summary(REML_modelg))
print(confint(REML_modelg))
## outputs Alex:
model_n <- REML_model$dim$N
ss <- summary(REML_model)
model_AIC <- ss$AIC
model_BIC <- ss$BIC
print("Below is the output requested by Alexander Braun")
print(round(t(data.frame(model_intercept=model_intercept,
                        model_pre_slope=model_pre_slope,
                        model_level_change=model_level_change,
                        model_slope_change=model_slope_change,
                        model_AIC=model_AIC,
                        model_BIC=model_BIC,
                        model_n=model_n)),5))

#####
### Create a plot ###
#####

```

```

dat0$prediction <- predict(REML_model)

# create the counterfactual data, recall this is what would have happened had there been no interruption
# and therefore is simple to generate by just using the intercept and pre_slope data
dat0$counterfactual <- model_intercept + model_pre_slope*dat0$Triage_Begin_Month

# for the graph we need the interruption time
int_time <- as.Date(x=intday,"%d%b%Y")

# colourblind colour blue
myblue <- rgb(0,114,178,max=255,alpha=255)
myothe <- rgb(248,118,109,max=255,alpha=255)

# legends can be tricky, I found this useful... https://stackoverflow.com/questions/17148679/construct-a-manual-legend
# set the colours to be used
colors <- c("Trend lines"=myblue,"Counterfactual"=myothe,"Data points"=myblue)
# set the line types (the lty values)
line_types <- c("Trend lines"=1,"Counterfactual"=2)

# set up the plot
ITS_plot <- ggplot(data=dat0,aes(x=Triage_Begin_Date,y=!sym(outcome))) +
  # add the actual data points
  geom_point(color="black", pch=1, size=1, stroke=1) +
  # set the predicted line in the pre-interruption segment
  geom_line(data=subset(dat0,Level_Change==0), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Trend lines")) +
  # set the predicted line in the post-interruption segment
  geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Trend lines")) +
  # set the counterfactual line in the post-interruption segment
  geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=counterfactual,color="Counterfactual",linetype="Counterfactual")) +
  # add some labels to the axes
  labs(x="Date",y="Number of Admissions per Month",title=ptitle) +

```

```

scale_y_continuous(limits = c(0, 2200)) +
# set the colours to be used using the vector we set up above
scale_color_manual(name="Legend:",values=c(myothe,myblue)) +
# set up the line types using the vector we set up above
scale_linetype_manual(name="Legend:",values=line_types) +
# add a vertical line for the interruption time
geom_vline(xintercept=int_time, color="black", lty="dashed", lwd=2) +
theme(legend.position = "bottom",
      #legend.background = element_rect(fill="white",linewidth=1,linetype="solid"),
      #legend.title = element_text(size=15),
      #legend.text = element_text(size=15),
      #legend.key.width = unit(4,"cm"),
      #legend.key = element_rect(colour=NA, fill=NA),
      #panel.background = element_blank(),
      #panel.grid.major = element_line(colour="gray"),
      #axis.title.y = element_text(face="bold", size=20),
      #axis.title.x = element_text(face="bold", size=20),
      #plot.title = element_text(face="bold", size=20),
      #axis.text.x=element_text(face="bold", size=15),
      #axis.text.y=element_text(face="bold", size=15),

)
# show this plot in the plots window
ITS_plot
}

###
getITSresultsInf <- function(dat0,outcome,ptitle,intday="01Jan2024"){

```

```

form <- formula(paste0(outcome, " ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza", sep=""))
print(form)
REML_model <- gls(model=form,
                  method="REML",
                  corr=corAR1(form=~Triage_Begin_Day),
                  #corr=corARMA(form=~Triage_Begin_Day,p=1,q=1),
                  data=dat0
)
# print the results of the analysis to the console
print(summary(REML_model)) # display the parameter estimates

# to save the parameter estimates:
# predicted values
model_estimates <- REML_model$fitted
# save the parameter estimates for display in graphs etc.
model_intercept <- REML_model$coefficients[1]
model_pre_slope <- REML_model$coefficients[2]
model_level_change <- REML_model$coefficients[3]
model_slope_change <- REML_model$coefficients[4]
model_influenza <- REML_model$coefficients[5]
# generate the confidence intervals
model_CIs <- confint(REML_model)
# display the confidence intervals in the console
print("Below are the model estimates together with their 95% confidence limits.")
print(cbind(`Point estimates`=c(model_intercept,
                                model_pre_slope,
                                model_level_change,
                                model_slope_change,
                                model_influenza),

            model_CIs))

###

```

```

print("Below are the model estimates together with multiplicity adjusted p-values and confidence limits")
set.seed(12345)
REML_modelg <- glht(REML_model)
print(summary(REML_modelg))
print(confint(REML_modelg))
## outputs Alex:
model_n <- REML_model$dim$N
ss <- summary(REML_model)
model_AIC <- ss$AIC
model_BIC <- ss$BIC
print("Below is the output requested by Alexander Braun")
print(round(t(data.frame(model_intercept=model_intercept,
                        model_pre_slope=model_pre_slope,
                        model_level_change=model_level_change,
                        model_slope_change=model_slope_change,
                        model_influenza=model_influenza,
                        model_AIC=model_AIC,
                        model_BIC=model_BIC,
                        model_n=model_n)),5))

#####
### Create a plot ###
#####
dat0$prediction <- predict(REML_model)

# create the counterfactual data, recall this is what would have happened had there been no interruption
# and therefore is simple to generate by just using the intercept and pre_slope data
dat0$counterfactual <- model_intercept + model_pre_slope*dat0$Triage_Begin_Day + model_influenza*dat0$influenza

# for the graph we need the interruption time
int_time <- as.Date(x=intday,"%d%b%Y")

```

```

# colourblind colour blue
myblue <- rgb(0,114,178,max=255,alpha=255)
myothe <- rgb(248,118,109,max=255,alpha=255)

# legends can be tricky, I found this useful... https://stackoverflow.com/questions/17148679/construct-a-manual-legend
# set the colours to be used
colors <- c("Trend lines"=myblue,"Counterfactual"=myblue,"Data points"=myblue)
# set the line types (the lty values)
line_types <- c("Trend lines"=1,"Counterfactual"=2)

# set up the plot
ITS_plot <- ggplot(data=dat0,aes(x=Triage_Begin_Date,y=!!sym(outcome))) +
  # add the actual data points
  geom_point(color="black", pch=1, size=1, stroke=1) +
  # set the predicted line in the pre-interruption segment
  geom_line(data=subset(dat0,Level_Change==0), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Trend lines")) +
  # set the predicted line in the post-interruption segment
  geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Trend lines")) +
  # set the counterfactual line in the post-interruption segment
  geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=counterfactual,color="Counterfactual",linetype="Counterfactual")) +
  # add some labels to the axes
  labs(x="Date",y="Number of Admissions per Day",title=ptitle) +
  scale_y_continuous(limits = c(0, 120)) +
  # set the colours to be used using the vector we set up above
  scale_color_manual(name="Legend:",values=c(myothe,myblue)) +
  # set up the line types using the vector we set up above
  scale_linetype_manual(name="Legend:",values=line_types) +
  # add a vertical line for the interruption time
  geom_vline(xintercept=int_time, color="black", lty="dashed", lwd=2) +

```

```

    theme(legend.position = "bottom",
          #legend.background = element_rect(fill="white",linewidth=1,linetype="solid"),
          #legend.title = element_text(size=15),
          #legend.text = element_text(size=15),
          #legend.key.width = unit(4,"cm"),
          #legend.key = element_rect(colour=NA, fill=NA),
          #panel.background = element_blank(),
          #panel.grid.major = element_line(colour="gray"),
          #axis.title.y = element_text(face="bold", size=20),
          #axis.title.x = element_text(face="bold", size=20),
          #plot.title = element_text(face="bold", size=20),
          #axis.text.x=element_text(face="bold", size=15),
          #axis.text.y=element_text(face="bold", size=15),
    )
    # show this plot in the plots window
    ITS_plot
  }

getITSresultsNew <- function(dat0,outcome,ptitle,intday="01Jan2024"){
  form <- formula(paste0(outcome, " ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza + Holiday + WD2 + WD3
  print(form)
  REML_model <- gls(model=form,
                    method="REML",
                    corr=corAR1(form=~Triage_Begin_Day),
                    #corr=corARMA(form=~Triage_Begin_Day,p=1,q=1),
                    data=dat0
  )
  # print the results of the analysis to the console
  print(summary(REML_model)) # display the parameter estimates

```

```

# to save the parameter estimates:
# predicted values
model_estimates <- REML_model$fitted
# save the parameter estimates for display in graphs etc.
model_intercept <- REML_model$coefficients[1]
model_pre_slope <- REML_model$coefficients[2]
model_level_change <- REML_model$coefficients[3]
model_slope_change <- REML_model$coefficients[4]
model_influenza <- REML_model$coefficients[5]
model_holiday <- REML_model$coefficients[6]
model_WD2 <- REML_model$coefficients[7]
model_WD3 <- REML_model$coefficients[8]
model_WD4 <- REML_model$coefficients[9]
model_WD5 <- REML_model$coefficients[10]
model_WD6 <- REML_model$coefficients[11]
model_WD7 <- REML_model$coefficients[12]

# generate the confidence intervals
model_CIs <- confint(REML_model)
# display the confidence intervals in the console
print("Below are the model estimates together with their 95% confidence limits.")
print(cbind(`Point estimates`=c(model_intercept,
                                model_pre_slope,
                                model_level_change,
                                model_slope_change,
                                model_influenza,
                                model_holiday,
                                model_WD2,
                                model_WD3,
                                model_WD4,
                                model_WD5,

```



```

                                model_WD6,
                                model_WD7),

                                model_CIs))

###
print("Below are the model estimates together with multiplicity adjusted p-values and confidence limits")
set.seed(12345)
REML_modelg <- glht(REML_model)
print(summary(REML_modelg))
print(confint(REML_modelg))
## outputs Alex:
model_n    <- REML_model$dim$N
ss          <- summary(REML_model)
model_AIC  <- ss$AIC
model_BIC  <- ss$BIC
print("Below is the output requested by Alexander Braun")
print(round(t(data.frame(model_intercept=model_intercept,
                        model_pre_slope=model_pre_slope,
                        model_level_change=model_level_change,
                        model_slope_change=model_slope_change,
                        model_influenza=model_influenza,
                        model_holiday=model_holiday,
                        model_WD2=model_WD2,
                        model_WD3=model_WD3,
                        model_WD4=model_WD4,
                        model_WD5=model_WD5,
                        model_WD6=model_WD6,
                        model_WD7=model_WD7,
                        model_AIC=model_AIC,
                        model_BIC=model_BIC,
                        model_n=model_n)),5))

```

```
#####
### Create a plot ###
#####

dat0$prediction <- predict(REML_model)

# create the counterfactual data, recall this is what would have happened had there been no interruption
# and therefore is simple to generate by just using the intercept and pre_slope data
dat0$counterfactual <- model_intercept + model_pre_slope*dat0$Triage_Begin_Day +
  model_influenza*dat0$influenza + model_holiday*dat0$Holiday +
  model_WD2*dat0$WD2 +
  model_WD3*dat0$WD3 +
  model_WD4*dat0$WD4 +
  model_WD5*dat0$WD5 +
  model_WD6*dat0$WD6 +
  model_WD7*dat0$WD7

# for the graph we need the interruption time
int_time <- as.Date(x=intday,"%d%b%Y")

# colourblind colour blue
myblue <- rgb(0,114,178,max=255,alpha=255)
myothe <- rgb(248,118,109,max=255,alpha=255)

# legends can be tricky, I found this useful... https://stackoverflow.com/questions/17148679/construct-a-manual-legend
# set the colours to be used
colors <- c("Trend lines"=myblue,"Counterfactual"=myblue,"Data points"=myblue)
# set the line types (the lty values)
line_types <- c("Trend lines"=1,"Counterfactual"=2)

# set up the plot
ITS_plot <- ggplot(data=dat0,aes(x=Triage_Begin_Date,y=!!sym(outcome))) +
```

```

# add the actual data points
geom_point(color="black", pch=1, size=1, stroke=1) +
# set the predicted line in the pre-interruption segment
geom_line(data=subset(dat0,Level_Change==0), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Tr
# set the predicted line in the post-interruption segment
geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Tr
# set the counterfactual line in the post-interruption segment
geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=counterfactual,color="Counterfactual",linet
# add some labels to the axes
labs(x="Date",y="Number of Admissions per Day",title=ptitle) +
scale_y_continuous(limits = c(0, 120)) +
# set the colours to be used using the vector we set up above
scale_color_manual(name="Legend:",values=c(myothe,myblue)) +
# set up the line types using the vector we set up above
scale_linetype_manual(name="Legend:",values=line_types) +
# add a vertical line for the interruption time
geom_vline(xintercept=int_time, color="black", lty="dashed", lwd=2) +
theme(legend.position = "bottom",
      #legend.background = element_rect(fill="white",linewidth=1,linetype="solid"),
      #legend.title = element_text(size=15),
      #legend.text = element_text(size=15),
      #legend.key.width = unit(4,"cm"),
      #legend.key = element_rect(colour=NA, fill=NA),
      #panel.background = element_blank(),
      #panel.grid.major = element_line(colour="gray"),
      #axis.title.y = element_text(face="bold", size=20),
      #axis.title.x = element_text(face="bold", size=20),
      #plot.title = element_text(face="bold", size=20),
      #axis.text.x=element_text(face="bold", size=15),
      #axis.text.y=element_text(face="bold", size=15),
)

```

```

# show this plot in the plots window
ITS_plot
}

getITSresultsInfARMA11 <- function(dat0,outcome,ptitle){
  form <- formula(paste0(outcome, " ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza",sep=""))
  print(form)
  REML_model <- gls(model=form,
                    method="REML",
                    #corr=corAR1(form=~Triage_Begin_Day),
                    corr=corARMA(form=~Triage_Begin_Day,p=1,q=1),
                    data=dat0
  )
  # print the results of the analysis to the console
  print(summary(REML_model)) # display the parameter estimates

  # to save the parameter estimates:
  # predicted values
  model_estimates <- REML_model$fitted
  # save the parameter estimates for display in graphs etc.
  model_intercept <- REML_model$coefficients[1]
  model_pre_slope <- REML_model$coefficients[2]
  model_level_change <- REML_model$coefficients[3]
  model_slope_change <- REML_model$coefficients[4]
  model_influenza <- REML_model$coefficients[5]
  # generate the confidence intervals
  model_CIs <- confint(REML_model)
  # display the confidence intervals in the console
  print("Below are the model estimates together with their 95% confidence limits.")
}

```

```

print(cbind(`Point estimates`=c(model_intercept,
                                model_pre_slope,
                                model_level_change,
                                model_slope_change,
                                model_influenza),

            model_CIs))

###
print("Below are the model estimates together with multiplicity adjusted p-values and confidence limits")
set.seed(12345)
REML_modelg <- glht(REML_model)
print(summary(REML_modelg))
print(confint(REML_modelg))
## outputs Alex:
model_n <- REML_model$dim$N
ss <- summary(REML_model)
model_AIC <- ss$AIC
model_BIC <- ss$BIC
print("Below is the output requested by Alexander Braun")
print(round(t(data.frame(model_intercept=model_intercept,
                        model_pre_slope=model_pre_slope,
                        model_level_change=model_level_change,
                        model_slope_change=model_slope_change,
                        model_influenza=model_influenza,
                        model_AIC=model_AIC,
                        model_BIC=model_BIC,
                        model_n=model_n)),5))

#####
### Create a plot ###
#####
dat0$prediction <- predict(REML_model)

```

```

# create the counterfactual data, recall this is what would have happened had there been no interruption
# and therefore is simple to generate by just using the intercept and pre_slope data
dat0$counterfactual <- model_intercept + model_pre_slope*dat0$Triage_Begin_Day + model_influenza*dat0$influenza

# for the graph we need the interruption time
int_time <- as.Date(x="01Jan2024", "%d%b%Y")

# colourblind colour blue
myblue <- rgb(0,114,178,max=255,alpha=255)
myothe <- rgb(248,118,109,max=255,alpha=255)

# legends can be tricky, I found this useful... https://stackoverflow.com/questions/17148679/construct-a-manual-legend
# set the colours to be used
colors <- c("Trend lines"=myblue,"Counterfactual"=myblue,"Data points"=myblue)
# set the line types (the lty values)
line_types <- c("Trend lines"=1,"Counterfactual"=2)

# set up the plot
ITS_plot <- ggplot(data=dat0,aes(x=Triage_Begin_Date,y=!!sym(outcome))) +
  # add the actual data points
  geom_point(color="black", pch=1, size=1, stroke=1) +
  # set the predicted line in the pre-interruption segment
  geom_line(data=subset(dat0,Level_Change==0), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Trend lines")) +
  # set the predicted line in the post-interruption segment
  geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Trend lines")) +
  # set the counterfactual line in the post-interruption segment
  geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=counterfactual,color="Counterfactual",linetype="Counterfactual")) +
  # add some labels to the axes
  labs(x="Date",y="Number of Admissions per Day",title=ptitle) +

```

```

scale_y_continuous(limits = c(0, 120)) +
# set the colours to be used using the vector we set up above
scale_color_manual(name="Legend:",values=c(myothe,myblue)) +
# set up the line types using the vector we set up above
scale_linetype_manual(name="Legend:",values=line_types) +
# add a vertical line for the interruption time
geom_vline(xintercept=int_time, color="black", lty="dashed", lwd=2) +
theme(legend.position = "bottom",
      #legend.background = element_rect(fill="white",linewidth=1,linetype="solid"),
      #legend.title = element_text(size=15),
      #legend.text = element_text(size=15),
      #legend.key.width = unit(4,"cm"),
      #legend.key = element_rect(colour=NA, fill=NA),
      #panel.background = element_blank(),
      #panel.grid.major = element_line(colour="gray"),
      #axis.title.y = element_text(face="bold", size=20),
      #axis.title.x = element_text(face="bold", size=20),
      #plot.title = element_text(face="bold", size=20),
      #axis.text.x=element_text(face="bold", size=15),
      #axis.text.y=element_text(face="bold", size=15),
    )
# show this plot in the plots window
ITS_plot
}

getITSresultsInfAR2 <- function(dat0,outcome,ptitle){
  form <- formula(paste0(outcome, " ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza",sep=""))
  print(form)
  REML_model <- gls(model=form,

```

```

        method="REML",
        #corr=corAR1(form=~Triage_Begin_Day),
        corr=corARMA(form=~Triage_Begin_Day,p=2,q=0),
        data=dat0
    )
    # print the results of the analysis to the console
    print(summary(REML_model)) # display the parameter estimates

    # to save the parameter estimates:
    # predicted values
    model_estimates <- REML_model$fitted
    # save the parameter estimates for display in graphs etc.
    model_intercept <- REML_model$coefficients[1]
    model_pre_slope <- REML_model$coefficients[2]
    model_level_change <- REML_model$coefficients[3]
    model_slope_change <- REML_model$coefficients[4]
    model_influenza <- REML_model$coefficients[5]
    # generate the confidence intervals
    model_CIs <- confint(REML_model)
    # display the confidence intervals in the console
    print("Below are the model estimates together with their 95% confidence limits.")
    print(cbind(`Point estimates`=c(model_intercept,
                                    model_pre_slope,
                                    model_level_change,
                                    model_slope_change,
                                    model_influenza),

                model_CIs))

    ###
    print("Below are the model estimates together with multiplicity adjusted p-values and confidence limits")
    set.seed(12345)
    REML_modelg <- glht(REML_model)

```



```

print(summary(REML_modelg))
print(confint(REML_modelg))
## outputs Alex:
model_n    <- REML_model$dim$N
ss          <- summary(REML_model)
model_AIC  <- ss$AIC
model_BIC  <- ss$BIC
print("Below is the output requested by Alexander Braun")
print(round(t(data.frame(model_intercept=model_intercept,
                        model_pre_slope=model_pre_slope,
                        model_level_change=model_level_change,
                        model_slope_change=model_slope_change,
                        model_influenza=model_influenza,
                        model_AIC=model_AIC,
                        model_BIC=model_BIC,
                        model_n=model_n)),5))

#####
### Create a plot ###
#####
dat0$prediction <- predict(REML_model)

# create the counterfactual data, recall this is what would have happened had there been no interruption
# and therefore is simple to generate by just using the intercept and pre_slope data
dat0$counterfactual <- model_intercept + model_pre_slope*dat0$Triage_Begin_Day + model_influenza*dat0$influenza

# for the graph we need the interruption time
int_time <- as.Date(x="01Jan2024", "%d%b%Y")

# colourblind colour blue

```

```

myblue <- rgb(0,114,178,max=255,alpha=255)
myothe <- rgb(248,118,109,max=255,alpha=255)

# legends can be tricky, I found this useful... https://stackoverflow.com/questions/17148679/construct-a-manual-legend
# set the colours to be used
colors <- c("Trend lines"=myblue,"Counterfactual"=myblue,"Data points"=myblue)
# set the line types (the lty values)
line_types <- c("Trend lines"=1,"Counterfactual"=2)

# set up the plot
ITS_plot <- ggplot(data=dat0,aes(x=Triage_Begin_Date,y=!sym(outcome))) +
  # add the actual data points
  geom_point(color="black", pch=1, size=1, stroke=1) +
  # set the predicted line in the pre-interruption segment
  geom_line(data=subset(dat0,Level_Change==0), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Trend lines")) +
  # set the predicted line in the post-interruption segment
  geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=prediction,color="Trend lines",linetype="Trend lines")) +
  # set the counterfactual line in the post-interruption segment
  geom_line(data=subset(dat0,Level_Change==1), aes(x=Triage_Begin_Date,y=counterfactual,color="Counterfactual",linetype="Counterfactual")) +
  # add some labels to the axes
  labs(x="Date",y="Number of Admissions per Day",title=ptitle) +
  scale_y_continuous(limits = c(0, 120)) +
  # set the colours to be used using the vector we set up above
  scale_color_manual(name="Legend:",values=c(myothe,myblue)) +
  # set up the line types using the vector we set up above
  scale_linetype_manual(name="Legend:",values=line_types) +
  # add a vertical line for the interruption time
  geom_vline(xintercept=int_time, color="black", lty="dashed", lwd=2) +
  theme(legend.position = "bottom",
        #legend.background = element_rect(fill="white",linewidth=1,linetype="solid"),
        #legend.title = element_text(size=15),

```

```

    #legend.text = element_text(size=15),
    #legend.key.width = unit(4,"cm"),
    #legend.key = element_rect(colour=NA, fill=NA),
    #panel.background = element_blank(),
    #panel.grid.major = element_line(colour="gray"),
    #axis.title.y = element_text(face="bold", size=20),
    #axis.title.x = element_text(face="bold", size=20),
    #plot.title = element_text(face="bold", size=20),
    #axis.text.x=element_text(face="bold", size=15),
    #axis.text.y=element_text(face="bold", size=15),
  )
  # show this plot in the plots window
  ITS_plot
}

###

getnts <- function(dat0,ptitle){
  est_n <- msmooth(dat0$n)
  plot(dat0$Triage_Begin_Date,est_n$orig,type="l",col="gray",
        las=1,lwd=1,
        main=str_c("Nonparametric Time Series Trend Estimation\n",ptitle),
        ylab="Number of Admissions per Day",
        ylim=c(0,120),
        xlim=c(as.Date(x="01Jan2022","%d%b%Y"),
               as.Date(x="31Dec2025","%d%b%Y")),
        xlab="Date")
  points(dat0$Triage_Begin_Date,est_n$ye,type="l",col="black",lwd=2)
  #points(cbest_n$np.estim$ye.ub,type="l",col="red",lwd=2)
  #points(cbest_n$np.estim$upper,type="l",col="red",lwd=1)
  #points(cbest_n$np.estim$lower,type="l",col="red",lwd=1)

```

```
abline(v=as.Date(x="01Jan2024", "%d%b%Y"), lwd=2, lty=2)
}
```

How do I perform the ITSA as suggested by Korevaar et al.?

```
getITSresults(dat0=datall, outcome="n", ptitle="All data")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change
```

```
<environment: 0x000002365b980708>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
      AIC      BIC    logLik
10980.94 11012.65 -5484.472
```

```
Correlation Structure: AR(1)
```

```
Formula: ~Triage_Begin_Day
```

```
Parameter estimate(s):
```

```
      Phi
0.2233564
```

```
Coefficients:
```

	Value	Std.Error	t-value	p-value
(Intercept)	48.04890	0.9791666	49.07122	0
Triage_Begin_Day	0.01312	0.0023201	5.65330	0
Level_Change	8.47466	1.3812324	6.13558	0
Slope_Change	-0.03278	0.0032803	-9.99245	0

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch
Triage_Begin_Day	-0.866		
Level_Change	0.354	-0.613	
Slope_Change	0.613	-0.708	0.002

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-3.67739615	-0.68004487	-0.05619068	0.58098473	6.00039220

Residual standard error: 10.53969

Degrees of freedom: 1461 total; 1457 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	48.04889937	46.129768071	49.96803067
Triage_Begin_Day	0.01311645	0.008569056	0.01766384
Level_Change	8.47465517	5.767489430	11.18182091
Slope_Change	-0.03277816	-0.039207418	-0.02634891

[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"

Simultaneous Tests for General Linear Hypotheses

Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	48.04890	0.97917	49.071	< 1e-09 ***
Triage_Begin_Day == 0	0.01312	0.00232	5.653	3.21e-08 ***
Level_Change == 0	8.47466	1.38123	6.136	2.06e-09 ***
Slope_Change == 0	-0.03278	0.00328	-9.992	< 1e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")

Quantile = 2.3979

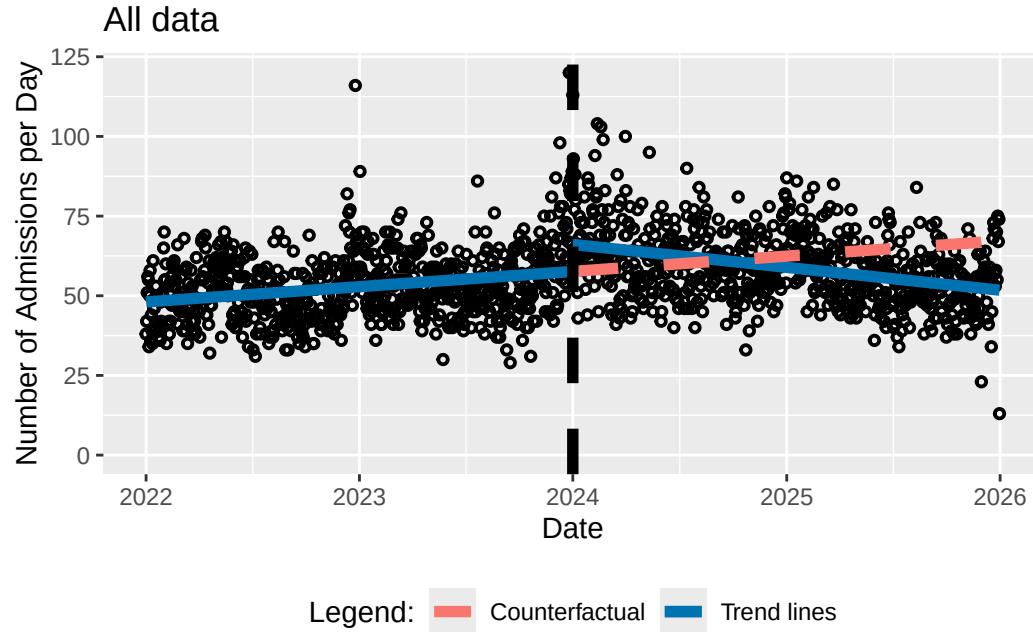
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	48.048899	45.701000	50.396798
Triage_Begin_Day == 0	0.013116	0.007553	0.018680
Level_Change == 0	8.474655	5.162661	11.786649
Slope_Change == 0	-0.032778	-0.040644	-0.024912

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	48.04890
model_pre_slope	0.01312
model_level_change	8.47466
model_slope_change	-0.03278
model_AIC	10980.94304
model_BIC	11012.64785
model_n	1461.00000



How do I perform the ITSA accounting for influenza?

```
getITSresultsInf(dat0=datall,outcome="n",ptitle="All data")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
<environment: 0x000002366317dd30>
Generalized least squares fit by REML
Model: form
Data: dat0
```

AIC	BIC	logLik
10922.29	10959.27	-5454.144

Correlation Structure: AR(1)

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi
0.1726749

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	47.81174	0.9015244	53.03432	0
Triage_Begin_Day	0.00979	0.0021751	4.50321	0
Level_Change	8.01940	1.2726598	6.30129	0
Slope_Change	-0.02626	0.0031236	-8.40838	0
influenza	5.83470	0.7254561	8.04281	0

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.355	-0.593		
Slope_Change	0.584	-0.721	-0.009	
influenza	-0.033	-0.191	-0.042	0.258

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-4.28553309	-0.70188619	-0.05811215	0.62930950	5.79696780

Residual standard error: 10.22125

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	47.811738071	46.04478269	49.57869345
Triage_Begin_Day	0.009794937	0.00553182	0.01405805
Level_Change	8.019395078	5.52502763	10.51376253
Slope_Change	-0.026264050	-0.03238611	-0.02014199
influenza	5.834704615	4.41283675	7.25657248

[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"

Simultaneous Tests for General Linear Hypotheses

Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	47.811738	0.901524	53.034	<1e-04 ***
Triage_Begin_Day == 0	0.009795	0.002175	4.503	<1e-04 ***
Level_Change == 0	8.019395	1.272660	6.301	<1e-04 ***
Slope_Change == 0	-0.026264	0.003124	-8.408	<1e-04 ***
influenza == 0	5.834705	0.725456	8.043	<1e-04 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")

Quantile = 2.5061

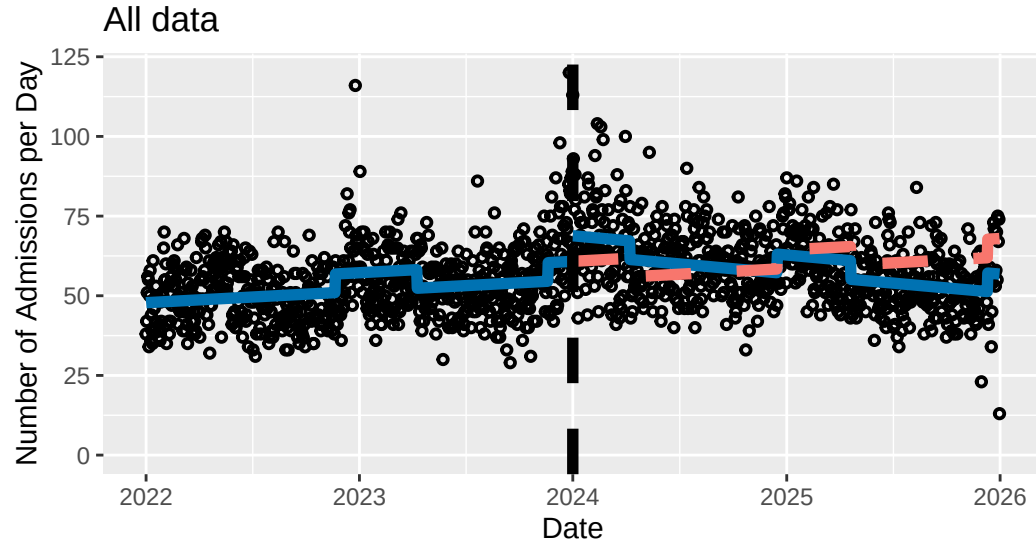
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	47.811738	45.552406	50.071070
Triage_Begin_Day == 0	0.009795	0.004344	0.015246
Level_Change == 0	8.019395	4.829952	11.208838
Slope_Change == 0	-0.026264	-0.034092	-0.018436
influenza == 0	5.834705	4.016622	7.652787

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	47.81174
model_pre_slope	0.00979
model_level_change	8.01940
model_slope_change	-0.02626
model_influenza	5.83470
model_AIC	10922.28704
model_BIC	10959.27118
model_n	1461.00000



Legend: Counterfactual Trend lines

```
getITSresultsInf(dat0=datsem,outcome="n",ptitle="Men only")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x000002365ff2a9e8>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
AIC      BIC      logLik
```

```
9573.872 9610.856 -4779.936
```

```
Correlation Structure: AR(1)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.1032389

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	22.949168	0.5234891	43.83887	0e+00
Triage_Begin_Day	0.004630	0.0012632	3.66509	3e-04
Level_Change	3.647379	0.7392968	4.93358	0e+00
Slope_Change	-0.012658	0.0018138	-6.97885	0e+00
influenza	2.997797	0.4218099	7.10699	0e+00

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.355	-0.593		
Slope_Change	0.584	-0.721	-0.009	
influenza	-0.033	-0.191	-0.042	0.258

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-3.47302619	-0.65573118	-0.08181918	0.57854863	4.51761715

Residual standard error: 6.368527

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	22.949168199	21.923148496	23.975187902
Triage_Begin_Day	0.004629712	0.002153904	0.007105520
Level_Change	3.647378838	2.198383741	5.096373935

```
Slope_Change      -0.012657934 -0.016212832 -0.009103036
influenza          2.997797287  2.171065157  3.824529417
```

[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	22.949168	0.523489	43.839	< 0.001 ***
Triage_Begin_Day == 0	0.004630	0.001263	3.665	0.00114 **
Level_Change == 0	3.647379	0.739297	4.934	< 0.001 ***
Slope_Change == 0	-0.012658	0.001814	-6.979	< 0.001 ***
influenza == 0	2.997797	0.421810	7.107	< 0.001 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Quantile = 2.5034

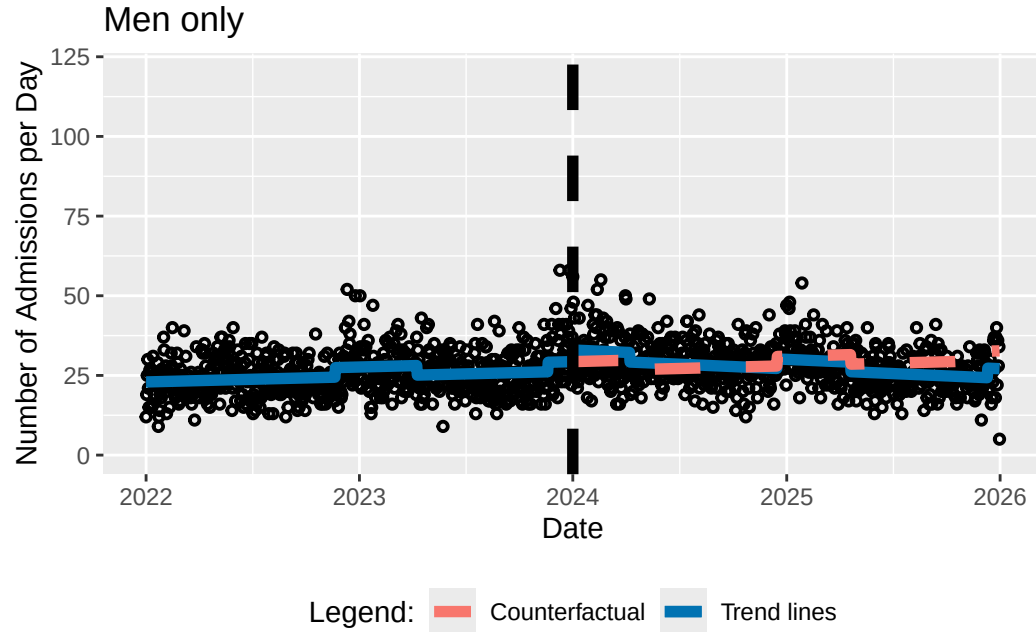
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	22.949168	21.638691	24.259645
Triage_Begin_Day == 0	0.004630	0.001468	0.007792
Level_Change == 0	3.647379	1.796659	5.498099
Slope_Change == 0	-0.012658	-0.017198	-0.008117
influenza == 0	2.997797	1.941859	4.053736

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	22.94917
model_pre_slope	0.00463
model_level_change	3.64738
model_slope_change	-0.01266
model_influenza	2.99780
model_AIC	9573.87173
model_BIC	9610.85587
model_n	1461.00000



```
getITSresultsInf(dat0=datsef,outcome="n",ptitle="Women only")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x000002365968cb70>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
AIC      BIC      logLik
```

```
9644.489 9681.474 -4815.245
```

```
Correlation Structure: AR(1)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.1280139

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	24.861788	0.5515659	45.07492	0e+00
Triage_Begin_Day	0.005166	0.0013309	3.88175	1e-04
Level_Change	4.342654	0.7788484	5.57574	0e+00
Slope_Change	-0.013537	0.0019110	-7.08366	0e+00
influenza	2.854636	0.4442439	6.42583	0e+00

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.355	-0.593		
Slope_Change	0.584	-0.721	-0.009	
influenza	-0.033	-0.191	-0.042	0.258

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-3.31956724	-0.67523459	-0.07500497	0.65882262	5.56639047

Residual standard error: 6.544443

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	24.861788109	23.780738841	25.942837376
Triage_Begin_Day	0.005166154	0.002557672	0.007774637
Level_Change	4.342653903	2.816139047	5.869168759


```
Slope_Change      -0.013537137 -0.017282701 -0.009791573
influenza          2.854636209  1.983934169  3.725338249
```

```
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"
```

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	24.861788	0.551566	45.075	< 1e-04 ***
Triage_Begin_Day == 0	0.005166	0.001331	3.882	0.000442 ***
Level_Change == 0	4.342654	0.778848	5.576	< 1e-04 ***
Slope_Change == 0	-0.013537	0.001911	-7.084	< 1e-04 ***
influenza == 0	2.854636	0.444244	6.426	< 1e-04 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Adjusted p values reported -- single-step method)
```

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

```
Quantile = 2.5034
```

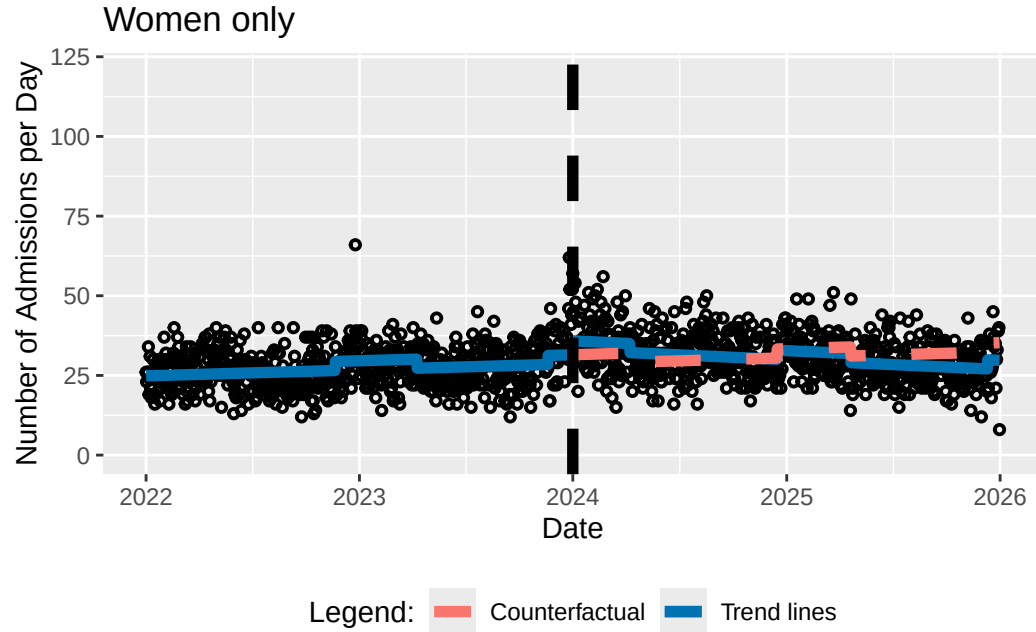
```
95% family-wise confidence level
```

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	24.861788	23.481024	26.242552
Triage_Begin_Day == 0	0.005166	0.001834	0.008498
Level_Change == 0	4.342654	2.392921	6.292387
Slope_Change == 0	-0.013537	-0.018321	-0.008753
influenza == 0	2.854636	1.742537	3.966736

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	24.86179
model_pre_slope	0.00517
model_level_change	4.34265
model_slope_change	-0.01354
model_influenza	2.85464
model_AIC	9644.48940
model_BIC	9681.47353
model_n	1461.00000



```
getITSresultsInf(dat0=datag12,outcome="n",ptitle="Age Category 1 and 2")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x0000023659434b38>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
AIC      BIC    logLik
```

```
9103.361 9140.345 -4544.68
```

```
Correlation Structure: AR(1)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.3490115

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	10.995993	0.6131733	17.932930	0.0000
Triage_Begin_Day	0.003620	0.0014784	2.448513	0.0145
Level_Change	3.635928	0.8638114	4.209169	0.0000
Slope_Change	-0.018959	0.0021245	-8.924178	0.0000
influenza	2.620889	0.4903746	5.344667	0.0000

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.354	-0.592		
Slope_Change	0.584	-0.721	-0.008	
influenza	-0.032	-0.190	-0.041	0.255

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-2.58310735	-0.66683304	-0.02697302	0.56935990	4.99538837

Residual standard error: 5.757066

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	10.995993362	9.7941958484	12.197790876
Triage_Begin_Day	0.003619773	0.0007222492	0.006517297
Level_Change	3.635928154	1.9428888345	5.328967474

```
Slope_Change      -0.018959228 -0.0231231304 -0.014795326
influenza          2.620888589  1.6597720890  3.582005089
```

```
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"
```

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	10.995993	0.613173	17.933	<0.001 ***
Triage_Begin_Day == 0	0.003620	0.001478	2.449	0.0578 .
Level_Change == 0	3.635928	0.863811	4.209	<0.001 ***
Slope_Change == 0	-0.018959	0.002124	-8.924	<0.001 ***
influenza == 0	2.620889	0.490375	5.345	<0.001 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Adjusted p values reported -- single-step method)
```

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

```
Quantile = 2.5024
```

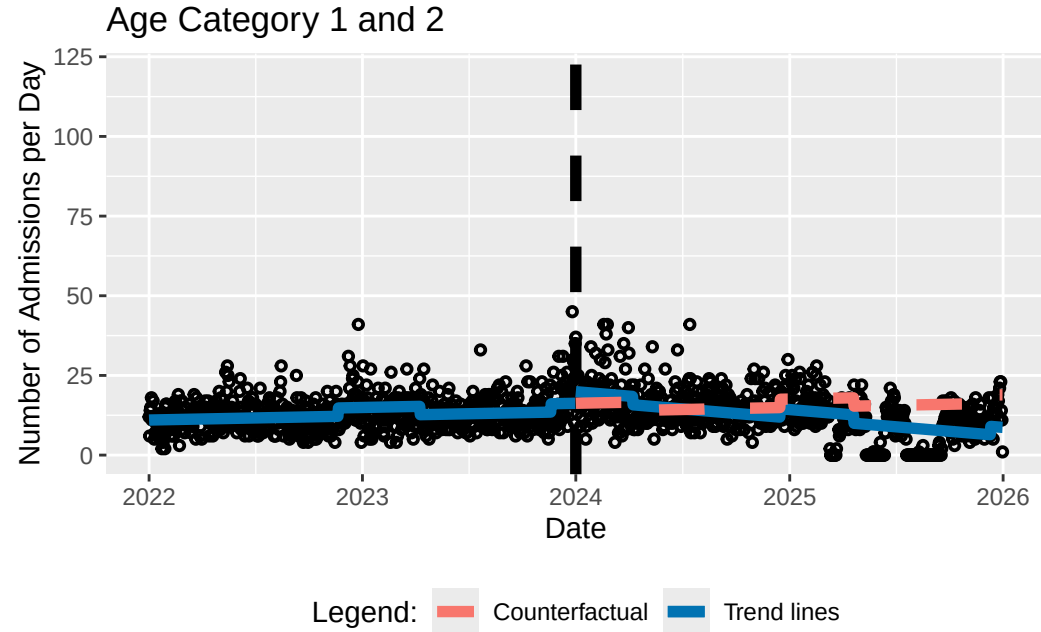
```
95% family-wise confidence level
```

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	1.100e+01	9.462e+00	1.253e+01
Triage_Begin_Day == 0	3.620e-03	-7.962e-05	7.319e-03
Level_Change == 0	3.636e+00	1.474e+00	5.798e+00
Slope_Change == 0	-1.896e-02	-2.428e-02	-1.364e-02
influenza == 0	2.621e+00	1.394e+00	3.848e+00

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	10.99599
model_pre_slope	0.00362
model_level_change	3.63593
model_slope_change	-0.01896
model_influenza	2.62089
model_AIC	9103.36068
model_BIC	9140.34482
model_n	1461.00000



```
getITSresultsInf(dat0=datag3,outcome="n",ptitle="Age Category 3")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x0000023662152df0>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
      AIC      BIC    logLik
```

```
9003.526 9040.51 -4494.763
```

```
Correlation Structure: AR(1)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.2883483

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	14.227095	0.5421346	26.242739	0.0000
Triage_Begin_Day	0.002984	0.0013075	2.282034	0.0226
Level_Change	1.951099	0.7644601	2.552258	0.0108
Slope_Change	-0.001271	0.0018784	-0.676911	0.4986
influenza	0.904311	0.4347622	2.080012	0.0377

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.354	-0.593		
Slope_Change	0.584	-0.721	-0.008	
influenza	-0.032	-0.190	-0.041	0.256

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-3.58514333	-0.68718827	-0.07010336	0.57594474	5.69046348

Residual standard error: 5.44291

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	14.227095340	13.1645311460	15.289659534
Triage_Begin_Day	0.002983769	0.0004211082	0.005546431
Level_Change	1.951098953	0.4527847445	3.449413162


```
Slope_Change      -0.001271481 -0.0049529924  0.002410031
influenza          0.904310626  0.0521924341  1.756428818
```

```
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"
```

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	14.227095	0.542135	26.243	<0.001 ***
Triage_Begin_Day == 0	0.002984	0.001308	2.282	0.0878 .
Level_Change == 0	1.951099	0.764460	2.552	0.0442 *
Slope_Change == 0	-0.001271	0.001878	-0.677	0.9232
influenza == 0	0.904311	0.434762	2.080	0.1406

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Quantile = 2.5016

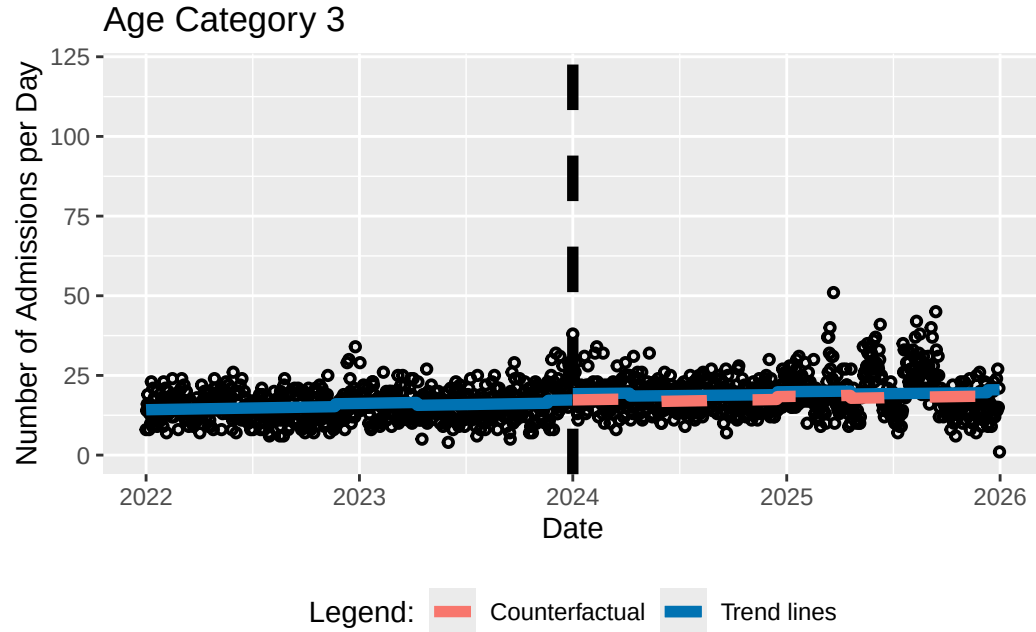
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	14.2270953	12.8708828	15.5833079
Triage_Begin_Day == 0	0.0029838	-0.0002871	0.0062546
Level_Change == 0	1.9510990	0.0387133	3.8634846
Slope_Change == 0	-0.0012715	-0.0059704	0.0034274
influenza == 0	0.9043106	-0.1832974	1.9919187

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	14.22710
model_pre_slope	0.00298
model_level_change	1.95110
model_slope_change	-0.00127
model_influenza	0.90431
model_AIC	9003.52554
model_BIC	9040.50968
model_n	1461.00000



```
getITSresultsInf(dat0=datag4,outcome="n",ptitle="Age Category 4")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x0000023658667ab0>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
AIC      BIC      logLik
```

```
8242.55 8279.534 -4114.275
```

```
Correlation Structure: AR(1)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.03429678

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	12.017288	0.3077530	39.04848	0.0000
Triage_Begin_Day	0.000840	0.0007427	1.13103	0.2582
Level_Change	1.141565	0.4347456	2.62582	0.0087
Slope_Change	-0.001458	0.0010663	-1.36747	0.1717
influenza	1.378943	0.2482173	5.55539	0.0000

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.356	-0.593		
Slope_Change	0.584	-0.721	-0.009	
influenza	-0.033	-0.191	-0.042	0.259

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-2.42461828	-0.67459537	-0.05624045	0.60010279	4.20733371

Residual standard error: 4.011498

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	12.0172883018	11.4141034267	1.262047e+01
Triage_Begin_Day	0.0008400006	-0.0006156346	2.295636e-03
Level_Change	1.1415648007	0.2894791645	1.993650e+00

```

Slope_Change      -0.0014581063 -0.0035479842 6.317715e-04
influenza          1.3789433150  0.8924463715 1.865440e+00
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"

```

Simultaneous Tests for General Linear Hypotheses

```

Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")

```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept) == 0	12.0172883	0.3077530	39.048	<0.001	***
Triage_Begin_Day == 0	0.0008400	0.0007427	1.131	0.6682	
Level_Change == 0	1.1415648	0.4347456	2.626	0.0359	*
Slope_Change == 0	-0.0014581	0.0010663	-1.367	0.5042	
influenza == 0	1.3789433	0.2482173	5.555	<0.001	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

```

Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")

```

Quantile = 2.5038

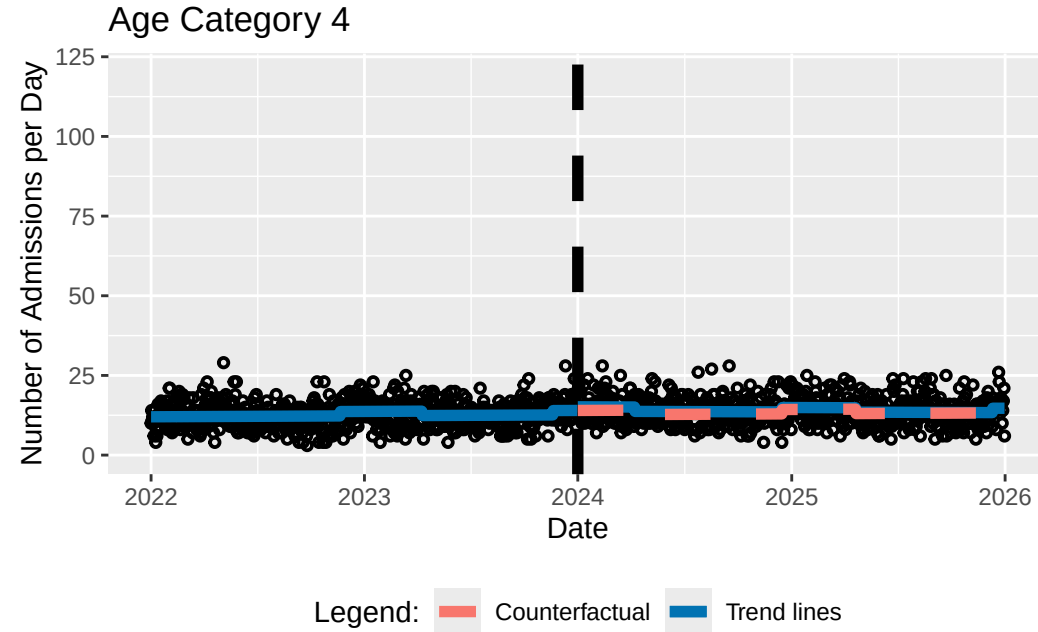
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	12.017288	11.246742	12.787835
Triage_Begin_Day == 0	0.000840	-0.001020	0.002700
Level_Change == 0	1.141565	0.053057	2.230073
Slope_Change == 0	-0.001458	-0.004128	0.001212
influenza == 0	1.378943	0.757462	2.000425

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	12.01729
model_pre_slope	0.00084
model_level_change	1.14156
model_slope_change	-0.00146
model_influenza	1.37894
model_AIC	8242.55007
model_BIC	8279.53420
model_n	1461.00000



```
getITSresultsInf(dat0=datag5,outcome="n",ptitle="Age Category 5")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x000002366186e530>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
AIC      BIC      logLik
```

```
8174.376 8211.36 -4080.188
```

```
Correlation Structure: AR(1)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.05729748

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	10.582233	0.3079646	34.36185	0.0000
Triage_Begin_Day	0.002388	0.0007432	3.21281	0.0013
Level_Change	1.338158	0.4350086	3.07616	0.0021
Slope_Change	-0.004744	0.0010670	-4.44620	0.0000
influenza	0.818100	0.2483155	3.29460	0.0010

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.355	-0.593		
Slope_Change	0.584	-0.721	-0.009	
influenza	-0.033	-0.191	-0.042	0.258

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-2.55372398	-0.71796306	-0.05003809	0.62792232	3.52861221

Residual standard error: 3.923133

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	10.582233360	9.978633899	11.185832820
Triage_Begin_Day	0.002387682	0.000931086	0.003844277
Level_Change	1.338157659	0.485556549	2.190758770


```
Slope_Change      -0.004744172 -0.006835488 -0.002652856
influenza          0.818100053  0.331410589  1.304789516
```

```
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"
```

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	10.5822334	0.3079646	34.362	< 0.001 ***
Triage_Begin_Day == 0	0.0023877	0.0007432	3.213	0.00619 **
Level_Change == 0	1.3381577	0.4350086	3.076	0.00923 **
Slope_Change == 0	-0.0047442	0.0010670	-4.446	< 0.001 ***
influenza == 0	0.8181001	0.2483155	3.295	0.00412 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Quantile = 2.5019

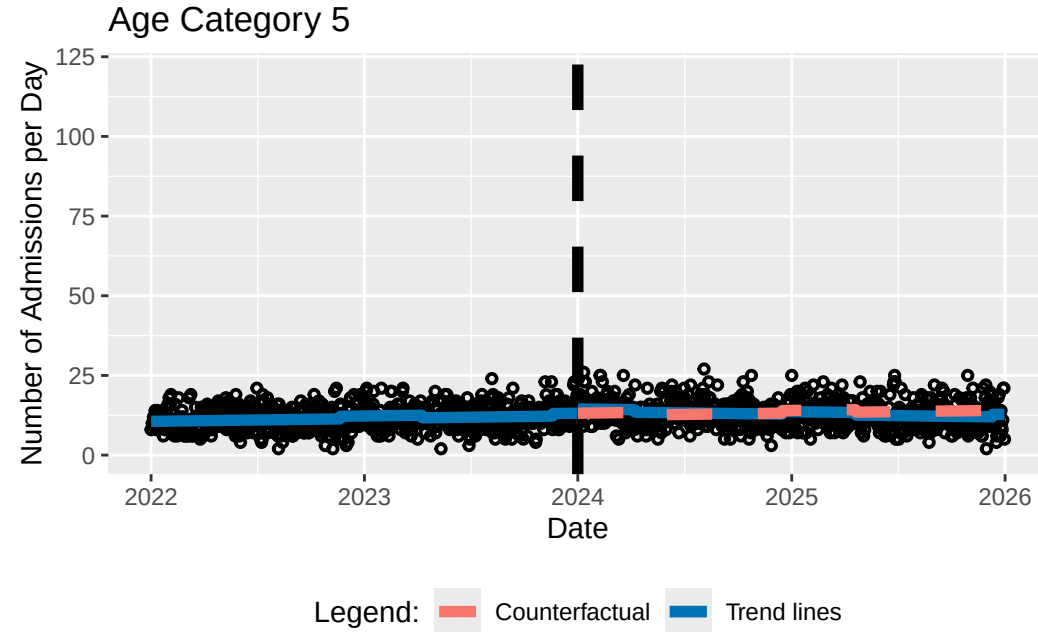
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	10.5822334	9.8117305	11.3527362
Triage_Begin_Day == 0	0.0023877	0.0005283	0.0042470
Level_Change == 0	1.3381577	0.2498009	2.4265145
Slope_Change == 0	-0.0047442	-0.0074138	-0.0020746
influenza == 0	0.8181001	0.1968344	1.4393657

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	10.58223
model_pre_slope	0.00239
model_level_change	1.33816
model_slope_change	-0.00474
model_influenza	0.81810
model_AIC	8174.37583
model_BIC	8211.35997
model_n	1461.00000



```
getITSresultsInf(dat0=datpr12,outcome="n",ptitle="Prio 1 and 2")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x0000023659575e80>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
      AIC      BIC    logLik
```

```
7129.239 7166.223 -3557.62
```

```
Correlation Structure: AR(1)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.09275989

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	5.973988	0.22349811	26.729481	0.0000
Triage_Begin_Day	-0.000427	0.00053931	-0.791334	0.4289
Level_Change	1.010878	0.31565052	3.202522	0.0014
Slope_Change	-0.001780	0.00077436	-2.299135	0.0216
influenza	0.405898	0.18011726	2.253522	0.0244

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.355	-0.593		
Slope_Change	0.584	-0.721	-0.009	
influenza	-0.033	-0.191	-0.042	0.258

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-2.32485654	-0.70084576	-0.06407648	0.65127154	4.65892232

Residual standard error: 2.747769

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	5.9739884983	5.535940253	6.4120367434
Triage_Begin_Day	-0.0004267779	-0.001483816	0.0006302596
Level_Change	1.0108778487	0.392214189	1.6295415088

```
Slope_Change      -0.0017803677 -0.003298093 -0.0002626419
influenza          0.4058982736  0.052874937  0.7589216104
```

```
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"
```

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	5.9739885	0.2234981	26.729	< 0.001 ***
Triage_Begin_Day == 0	-0.0004268	0.0005393	-0.791	0.87408
Level_Change == 0	1.0108778	0.3156505	3.203	0.00545 **
Slope_Change == 0	-0.0017804	0.0007744	-2.299	0.08442 .
influenza == 0	0.4058983	0.1801173	2.254	0.09396 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Quantile = 2.5003

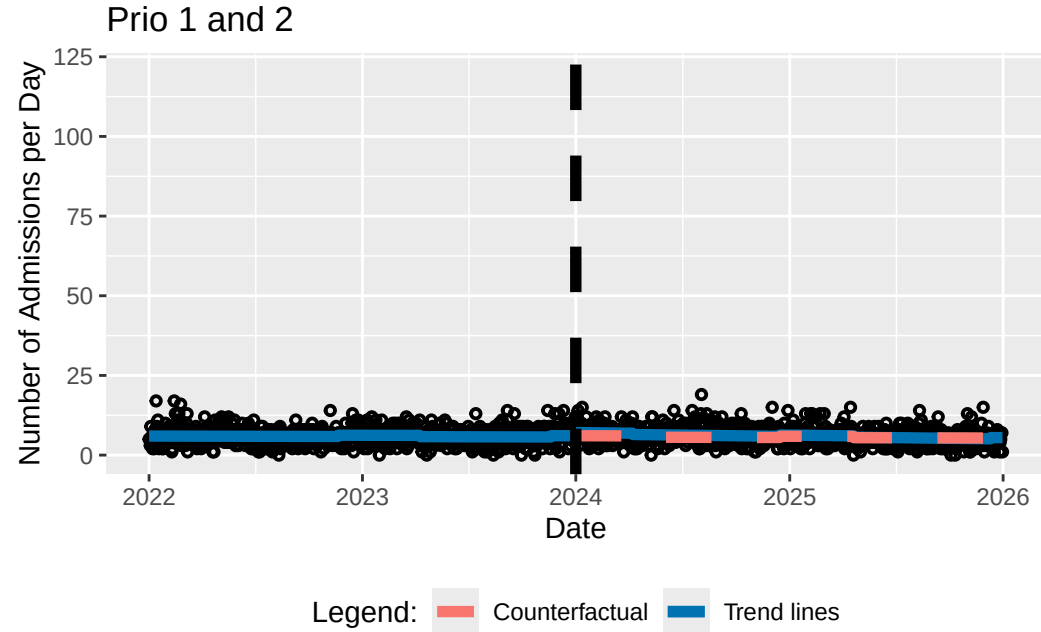
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	5.9739885	5.4151736	6.5328034
Triage_Begin_Day == 0	-0.0004268	-0.0017752	0.0009217
Level_Change == 0	1.0108778	0.2216532	1.8001025
Slope_Change == 0	-0.0017804	-0.0037165	0.0001558
influenza == 0	0.4058983	-0.0444510	0.8562476

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	5.97399
model_pre_slope	-0.00043
model_level_change	1.01088
model_slope_change	-0.00178
model_influenza	0.40590
model_AIC	7129.23925
model_BIC	7166.22339
model_n	1461.00000



```
getITSresultsInf(dat0=datpr3,outcome="n",ptitle="Prio 3")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x00000236594ad510>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
AIC      BIC  logLik
```

```
9276.799 9313.783 -4631.4
```

```
Correlation Structure: AR(1)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.05089041

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	16.740935	0.4466579	37.48044	0e+00
Triage_Begin_Day	0.007981	0.0010779	7.40441	0e+00
Level_Change	2.396005	0.6309319	3.79756	2e-04
Slope_Change	-0.014166	0.0015476	-9.15386	0e+00
influenza	2.428350	0.3601759	6.74212	0e+00

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.356	-0.593		
Slope_Change	0.584	-0.721	-0.009	
influenza	-0.033	-0.191	-0.042	0.258

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-3.08177979	-0.67349996	-0.06219026	0.61097619	4.88355214

Residual standard error: 5.726472

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	16.740935374	15.865501895	17.61636885
Triage_Begin_Day	0.007981035	0.005868437	0.01009363
Level_Change	2.396004702	1.159400827	3.63260858


```
Slope_Change      -0.014166087 -0.017199237 -0.01113294
influenza          2.428349782  1.722417928  3.13428164
```

```
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"
```

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	16.740935	0.446658	37.480	< 1e-04 ***
Triage_Begin_Day == 0	0.007981	0.001078	7.404	< 1e-04 ***
Level_Change == 0	2.396005	0.630932	3.798	0.000605 ***
Slope_Change == 0	-0.014166	0.001548	-9.154	< 1e-04 ***
influenza == 0	2.428350	0.360176	6.742	< 1e-04 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Adjusted p values reported -- single-step method)
```

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

```
Quantile = 2.5061
```

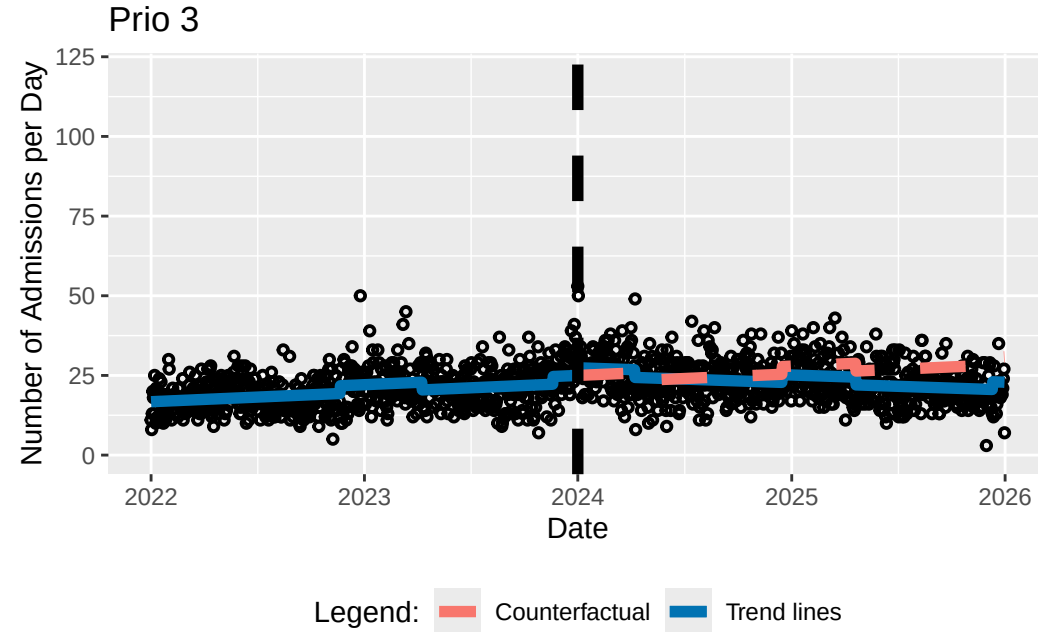
```
95% family-wise confidence level
```

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	16.740935	15.621560	17.860311
Triage_Begin_Day == 0	0.007981	0.005280	0.010682
Level_Change == 0	2.396005	0.814818	3.977191
Slope_Change == 0	-0.014166	-0.018044	-0.010288
influenza == 0	2.428350	1.525708	3.330991

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	16.74094
model_pre_slope	0.00798
model_level_change	2.39600
model_slope_change	-0.01417
model_influenza	2.42835
model_AIC	9276.79923
model_BIC	9313.78337
model_n	1461.00000



```
getITSresultsInf(dat0=datpr45,outcome="n",ptitle="Prio 4 and 5")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
<environment: 0x0000023661237728>
```

Generalized least squares fit by REML

Model: form

Data: dat0

AIC	BIC	logLik
9911.556	9948.541	-4948.778

Correlation Structure: AR(1)

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.1372317

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	25.090692	0.6109968	41.06518	0.0000
Triage_Begin_Day	0.002259	0.0014743	1.53210	0.1257
Level_Change	4.557025	0.8627239	5.28214	0.0000
Slope_Change	-0.010228	0.0021170	-4.83154	0.0000
influenza	3.027693	0.4920268	6.15351	0.0000

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.355	-0.593		
Slope_Change	0.584	-0.721	-0.009	
influenza	-0.033	-0.191	-0.042	0.258

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-3.27319930	-0.66000725	-0.07843921	0.58310574	6.02098622

Residual standard error: 7.182221

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	25.090692118	23.8931604155	26.288223821
Triage_Begin_Day	0.002258705	-0.0006307893	0.005148199
Level_Change	4.557025327	2.8661175122	6.247933142

```
Slope_Change      -0.010228133 -0.0143772800 -0.006078986
influenza          3.027692635  2.0633378268  3.992047443
```

```
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"
```

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	25.090692	0.610997	41.065	<0.001 ***
Triage_Begin_Day == 0	0.002259	0.001474	1.532	0.397
Level_Change == 0	4.557025	0.862724	5.282	<0.001 ***
Slope_Change == 0	-0.010228	0.002117	-4.832	<0.001 ***
influenza == 0	3.027693	0.492027	6.154	<0.001 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Quantile = 2.5034

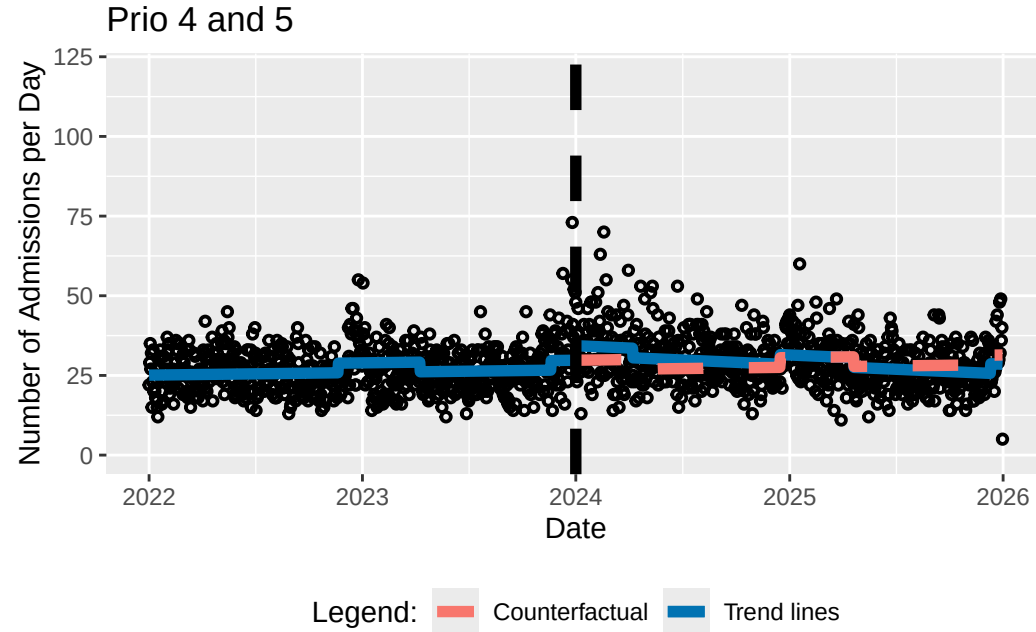
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	25.090692	23.561151	26.620233
Triage_Begin_Day == 0	0.002259	-0.001432	0.005949
Level_Change == 0	4.557025	2.397322	6.716729
Slope_Change == 0	-0.010228	-0.015528	-0.004929
influenza == 0	3.027693	1.795975	4.259410

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	25.09069
model_pre_slope	0.00226
model_level_change	4.55703
model_slope_change	-0.01023
model_influenza	3.02769
model_AIC	9911.55638
model_BIC	9948.54052
model_n	1461.00000



```
getITSresultsInf(dat0=datwk,outcome="n",ptitle="Weekdays (Monday-Friday) only")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x00000236585088b8>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
AIC      BIC      logLik
```

```
7720.332 7754.947 -3853.166
```

```
Correlation Structure: ARMA(1,0)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi1

0.1880405

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	49.78810	1.0010615	49.73531	0.0000
Triage_Begin_Day	0.00656	0.0024169	2.71535	0.0067
Level_Change	7.11975	1.4096253	5.05081	0.0000
Slope_Change	-0.01995	0.0034576	-5.76944	0.0000
influenza	5.09789	0.8039358	6.34116	0.0000

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.358	-0.596		
Slope_Change	0.586	-0.723	-0.002	
influenza	-0.033	-0.190	-0.042	0.258

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-4.48830225	-0.70512083	-0.05332372	0.57280632	6.15234772

Residual standard error: 9.81025

Degrees of freedom: 1043 total; 1038 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	49.788099516	47.826055099	51.75014393
Triage_Begin_Day	0.006562674	0.001825682	0.01129967
Level_Change	7.119754542	4.356939652	9.88256943


```
Slope_Change      -0.019948550 -0.026725362 -0.01317174
influenza          5.097889527  3.522204276  6.67357478
```

[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	49.788100	1.001061	49.735	<0.001 ***
Triage_Begin_Day == 0	0.006563	0.002417	2.715	0.028 *
Level_Change == 0	7.119755	1.409625	5.051	<0.001 ***
Slope_Change == 0	-0.019949	0.003458	-5.769	<0.001 ***
influenza == 0	5.097890	0.803936	6.341	<0.001 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Quantile = 2.5019

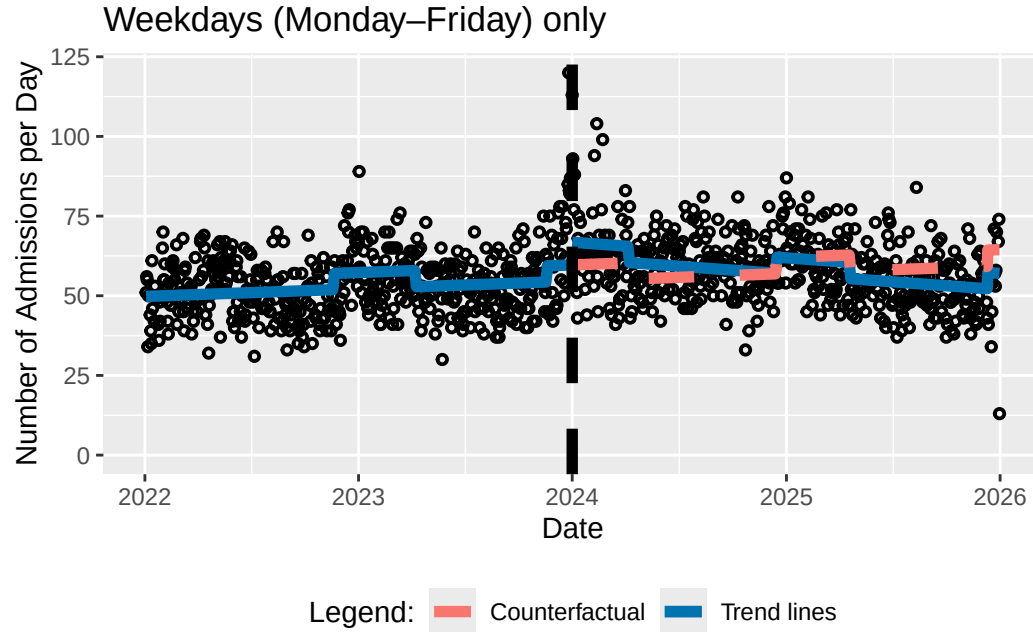
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	49.7880995	47.2835524	52.2926466
Triage_Begin_Day == 0	0.0065627	0.0005159	0.0126094
Level_Change == 0	7.1197545	3.5930250	10.6464841
Slope_Change == 0	-0.0199485	-0.0285991	-0.0112980
influenza == 0	5.0978895	3.0865294	7.1092497

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	49.78810
model_pre_slope	0.00656
model_level_change	7.11975
model_slope_change	-0.01995
model_influenza	5.09789
model_AIC	7720.33201
model_BIC	7754.94737
model_n	1043.00000



```
getITSresultsInf(dat0=datss,outcome="n",ptitle="Weekends (Saturday–Sunday) only")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x000002365e622a20>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
AIC      BIC      logLik
```

```
3200.457 3228.622 -1593.229
```

```
Correlation Structure: ARMA(1,0)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi1

0.08967834

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	43.10923	1.5576664	27.675518	0
Triage_Begin_Day	0.01832	0.0037550	4.878708	0
Level_Change	10.15772	2.2183475	4.578957	0
Slope_Change	-0.04211	0.0054333	-7.749587	0
influenza	7.88377	1.2678822	6.218064	0

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.842			
Level_Change	0.349	-0.587		
Slope_Change	0.578	-0.715	-0.024	
influenza	-0.031	-0.194	-0.043	0.259

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-2.65036143	-0.66473962	-0.08225554	0.63567037	5.38494756

Residual standard error: 10.85068

Degrees of freedom: 418 total; 413 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	43.10922516	40.0562551	46.16219520
Triage_Begin_Day	0.01831941	0.0109598	0.02567902
Level_Change	10.15771760	5.8098363	14.50559888

```
Slope_Change      -0.04210591 -0.0527550 -0.03145682
influenza          7.88377226  5.3987688 10.36877573
```

```
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"
```

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	43.109225	1.557666	27.676	< 1e-05 ***
Triage_Begin_Day == 0	0.018319	0.003755	4.879	< 1e-05 ***
Level_Change == 0	10.157718	2.218348	4.579	1.96e-05 ***
Slope_Change == 0	-0.042106	0.005433	-7.750	< 1e-05 ***
influenza == 0	7.883772	1.267882	6.218	< 1e-05 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Adjusted p values reported -- single-step method)
```

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

```
Quantile = 2.507
```

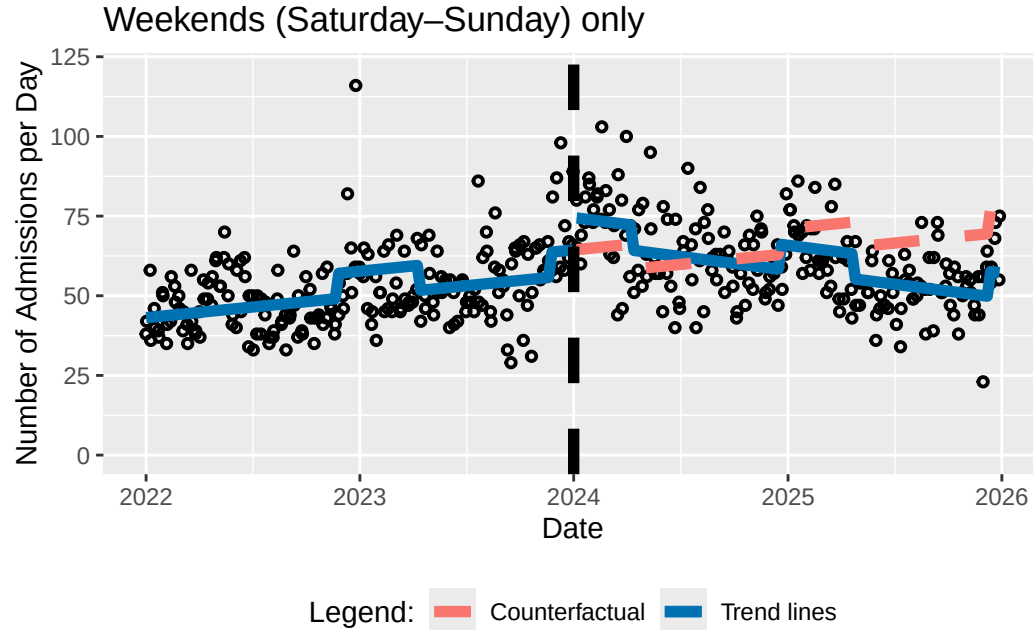
```
95% family-wise confidence level
```

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	43.109225	39.204106	47.014344
Triage_Begin_Day == 0	0.018319	0.008906	0.027733
Level_Change == 0	10.157718	4.596250	15.719185
Slope_Change == 0	-0.042106	-0.055727	-0.028484
influenza == 0	7.883772	4.705152	11.062393

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	43.10923
model_pre_slope	0.01832
model_level_change	10.15772
model_slope_change	-0.04211
model_influenza	7.88377
model_AIC	3200.45748
model_BIC	3228.62161
model_n	418.00000



How do I perform the ITSA accounting for influenza, holidays, and day of the week?

```
getITSresultsNew(dat0=datnew,outcome="n",ptitle="All data")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza +
  Holiday + WD2 + WD3 + WD4 + WD5 + WD6 + WD7
<environment: 0x000002365e9b2ed8>
Generalized least squares fit by REML
Model: form
```

Data: dat0
 AIC BIC logLik
 10823.22 10897.12 -5397.612

Correlation Structure: AR(1)
 Formula: ~Triage_Begin_Day
 Parameter estimate(s):
 Phi
 0.1612854

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	52.98309	1.0589001	50.03597	0.0000
Triage_Begin_Day	0.00981	0.0020763	4.72535	0.0000
Level_Change	8.01474	1.2149625	6.59670	0.0000
Slope_Change	-0.02629	0.0029816	-8.81861	0.0000
influenza	5.82106	0.6926832	8.40364	0.0000
Holiday	4.32481	1.3802088	3.13344	0.0018
WD2	-7.03539	0.8878655	-7.92393	0.0000
WD3	-7.46115	0.9552155	-7.81096	0.0000
WD4	-7.12872	0.9644035	-7.39185	0.0000
WD5	-5.71628	0.9665843	-5.91390	0.0000
WD6	-6.25334	0.9556991	-6.54321	0.0000
WD7	-3.69112	0.8867968	-4.16230	0.0000

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch	inflnz	Holiday	WD2	WD3	WD4
Triage_Begin_Day	-0.684								
Level_Change	0.286	-0.593							
Slope_Change	0.474	-0.721	-0.009						
influenza	-0.025	-0.191	-0.042	0.258					

Holiday	-0.089	-0.004	0.008	-0.003	-0.009					
WD2	-0.423	-0.001	0.002	0.000	-0.001	0.089				
WD3	-0.454	-0.002	0.003	0.000	-0.001	0.069	0.542			
WD4	-0.452	-0.003	0.003	0.002	-0.001	-0.001	0.468	0.572		
WD5	-0.457	-0.004	0.004	0.002	-0.001	0.067	0.462	0.507	0.576	
WD6	-0.455	-0.002	0.004	0.000	-0.001	0.076	0.456	0.492	0.503	
WD7	-0.422	-0.003	0.006	0.000	-0.001	0.075	0.423	0.455	0.457	

WD5 WD6

Triage_Begin_Day

Level_Change

Slope_Change

influenza

Holiday

WD2

WD3

WD4

WD5

WD6 0.576

WD7 0.473 0.541

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-4.20398044	-0.70170260	-0.05982397	0.63340118	5.75350732

Residual standard error: 9.871056

Degrees of freedom: 1461 total; 1449 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	52.983090992	50.907684993	55.05849699
Triage_Begin_Day	0.009811355	0.005741837	0.01388087
Level_Change	8.014737916	5.633455228	10.39602060

Slope_Change	-0.026293704	-0.032137566	-0.02044984
influenza	5.821063085	4.463428887	7.17869728
Holiday	4.324805329	1.619645878	7.02996478
WD2	-7.035386059	-8.775570435	-5.29520168
WD3	-7.461150192	-9.333338177	-5.58896221
WD4	-7.128724784	-9.018920944	-5.23852862
WD5	-5.716282011	-7.610752403	-3.82181162
WD6	-6.253342270	-8.126478045	-4.38020650
WD7	-3.691117400	-5.429207101	-1.95302770

[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	52.983091	1.058900	50.036	<0.001 ***
Triage_Begin_Day == 0	0.009811	0.002076	4.725	<0.001 ***
Level_Change == 0	8.014738	1.214962	6.597	<0.001 ***
Slope_Change == 0	-0.026294	0.002982	-8.819	<0.001 ***
influenza == 0	5.821063	0.692683	8.404	<0.001 ***
Holiday == 0	4.324805	1.380209	3.133	0.0182 *
WD2 == 0	-7.035386	0.887865	-7.924	<0.001 ***
WD3 == 0	-7.461150	0.955216	-7.811	<0.001 ***
WD4 == 0	-7.128725	0.964404	-7.392	<0.001 ***
WD5 == 0	-5.716282	0.966584	-5.914	<0.001 ***
WD6 == 0	-6.253342	0.955699	-6.543	<0.001 ***
WD7 == 0	-3.691117	0.886797	-4.162	<0.001 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
 (Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
 method = "REML")

Quantile = 2.8121
 95% family-wise confidence level

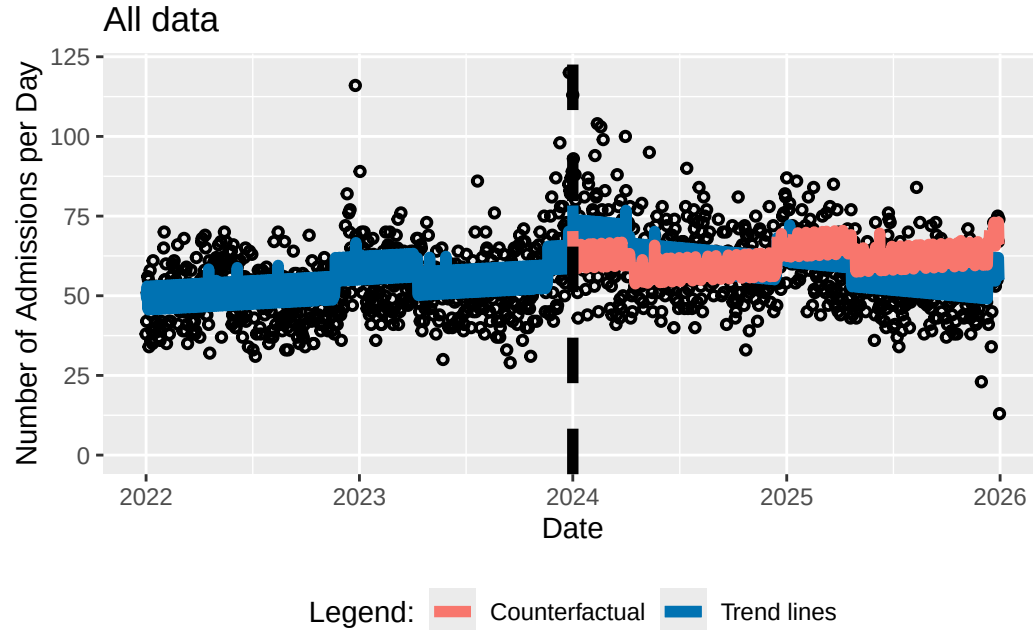
Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	52.983091	50.005325	55.960857
Triage_Begin_Day == 0	0.009811	0.003972	0.015650
Level_Change == 0	8.014738	4.598104	11.431372
Slope_Change == 0	-0.026294	-0.034678	-0.017909
influenza == 0	5.821063	3.873147	7.768980
Holiday == 0	4.324805	0.443477	8.206134
WD2 == 0	-7.035386	-9.532181	-4.538591
WD3 == 0	-7.461150	-10.147342	-4.774958
WD4 == 0	-7.128725	-9.840754	-4.416695
WD5 == 0	-5.716282	-8.434444	-2.998120
WD6 == 0	-6.253342	-8.940894	-3.565791
WD7 == 0	-3.691117	-6.184907	-1.197328

[1] "Below is the output requested by Alexander Braun"

(Intercept)
 model_intercept 52.98309

model_pre_slope	0.00981
model_level_change	8.01474
model_slope_change	-0.02629
model_influenza	5.82106
model_holiday	4.32481
model_WD2	-7.03539
model_WD3	-7.46115
model_WD4	-7.12872
model_WD5	-5.71628
model_WD6	-6.25334
model_WD7	-3.69112
model_AIC	10823.22376
model_BIC	10897.12457
model_n	1461.00000



How do I perform the ITSA accounting for influenza with different interruption dates?

```
getITSresultsInf(dat0=datalld,outcome="n",ptitle="All data",intday="01Dec2023")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
<environment: 0x0000023656169be0>
Generalized least squares fit by REML
Model: form
Data: dat0
```

AIC	BIC	logLik
10899.93	10936.92	-5442.967

Correlation Structure: AR(1)

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi
0.1533249

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	48.77453	0.8953534	54.47518	0.000
Triage_Begin_Day	0.00626	0.0022240	2.81403	0.005
Level_Change	10.85102	1.2504949	8.67738	0.000
Slope_Change	-0.02336	0.0030329	-7.70052	0.000
influenza	5.20095	0.7109355	7.31564	0.000

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.850			
Level_Change	0.368	-0.596		
Slope_Change	0.612	-0.752	0.034	
influenza	-0.074	-0.113	-0.160	0.242

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-4.27285448	-0.70015889	-0.04035678	0.63273820	5.91297248

Residual standard error: 10.10959

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	48.774534913	47.019674571	50.52939525
Triage_Begin_Day	0.006258451	0.001899456	0.01061745
Level_Change	10.851022296	8.400097327	13.30194727
Slope_Change	-0.023355173	-0.029299614	-0.01741073
influenza	5.200945836	3.807537793	6.59435388

[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"

Simultaneous Tests for General Linear Hypotheses

Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	48.774535	0.895353	54.475	<0.001 ***
Triage_Begin_Day == 0	0.006258	0.002224	2.814	0.0209 *
Level_Change == 0	10.851022	1.250495	8.677	<0.001 ***
Slope_Change == 0	-0.023355	0.003033	-7.701	<0.001 ***
influenza == 0	5.200946	0.710936	7.316	<0.001 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")

Quantile = 2.4988

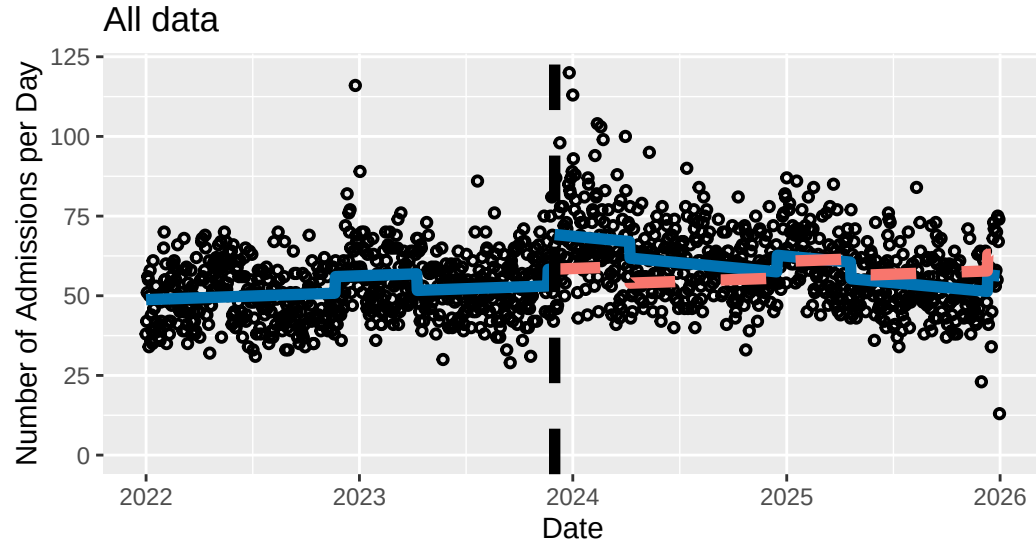
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	48.7745349	46.5372322	51.0118377
Triage_Begin_Day == 0	0.0062585	0.0007011	0.0118158
Level_Change == 0	10.8510223	7.7262943	13.9757502
Slope_Change == 0	-0.0233552	-0.0309338	-0.0157765
influenza == 0	5.2009458	3.4244651	6.9774266

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	48.77453
model_pre_slope	0.00626
model_level_change	10.85102
model_slope_change	-0.02336
model_influenza	5.20095
model_AIC	10899.93376
model_BIC	10936.91790
model_n	1461.00000



Legend: Counterfactual Trend lines

```
getITSresultsInf(dat0=datallf,outcome="n",ptitle="All data",intday="01Feb2024")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x0000023664d8ba70>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
AIC      BIC      logLik
```

```
10932.96 10969.95 -5459.481
```

```
Correlation Structure: AR(1)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.1799252

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	47.23603	0.8935582	52.86285	0
Triage_Begin_Day	0.01185	0.0021020	5.63681	0
Level_Change	6.02360	1.2920592	4.66202	0
Slope_Change	-0.02811	0.0031739	-8.85768	0
influenza	6.21688	0.7355956	8.45149	0

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.837			
Level_Change	0.346	-0.597		
Slope_Change	0.555	-0.684	-0.050	
influenza	0.000	-0.257	0.075	0.251

Standardized residuals:

Min	Q1	Med	Q3	Max
-4.2964532	-0.6863088	-0.0462820	0.6160953	5.6746134

Residual standard error: 10.27266

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	47.23602761	45.484685766	48.98736946
Triage_Begin_Day	0.01184874	0.007728834	0.01596864
Level_Change	6.02360172	3.491212237	8.55599119

```
Slope_Change      -0.02811374 -0.034334538 -0.02189293
influenza          6.21688098  4.775140118  7.65862185
```

[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	47.236028	0.893558	52.863	< 1e-05 ***
Triage_Begin_Day == 0	0.011849	0.002102	5.637	< 1e-05 ***
Level_Change == 0	6.023602	1.292059	4.662	1.32e-05 ***
Slope_Change == 0	-0.028114	0.003174	-8.858	< 1e-05 ***
influenza == 0	6.216881	0.735596	8.451	< 1e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Quantile = 2.509

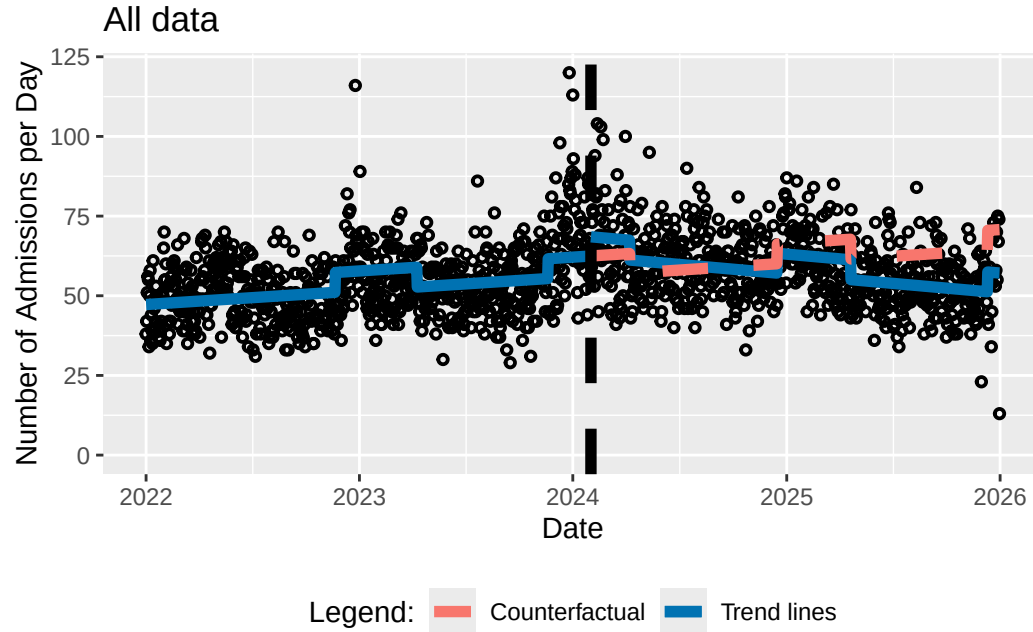
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	47.236028	44.994101	49.477954
Triage_Begin_Day == 0	0.011849	0.006575	0.017123
Level_Change == 0	6.023602	2.781841	9.265363
Slope_Change == 0	-0.028114	-0.036077	-0.020150
influenza == 0	6.216881	4.371281	8.062481

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	47.23603
model_pre_slope	0.01185
model_level_change	6.02360
model_slope_change	-0.02811
model_influenza	6.21688
model_AIC	10932.96281
model_BIC	10969.94695
model_n	1461.00000



```
getITSresultsInf(dat0=datall5,outcome="n",ptitle="All data",intday="01Jan2025")
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
<environment: 0x0000023659618c88>
Generalized least squares fit by REML
Model: form
Data: dat0
      AIC      BIC    logLik
10941.16 10978.14 -5463.579

Correlation Structure: AR(1)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

Phi

0.186204

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	47.01586	0.7658113	61.39353	0.000
Triage_Begin_Day	0.01282	0.0011940	10.73478	0.000
Level_Change	-4.02960	1.5415505	-2.61399	0.009
Slope_Change	-0.03145	0.0066122	-4.75692	0.000
influenza	6.65284	0.7537827	8.82594	0.000

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.830			
Level_Change	0.282	-0.398		
Slope_Change	0.097	-0.210	-0.675	
influenza	-0.184	-0.095	-0.230	0.308

Standardized residuals:

Min	Q1	Med	Q3	Max
-4.2560602	-0.6882021	-0.0608242	0.6138008	5.5949287

Residual standard error: 10.31828

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	47.01586037	45.51489772	48.51682303
Triage_Begin_Day	0.01281681	0.01047671	0.01515691
Level_Change	-4.02959905	-7.05098254	-1.00821555

```
Slope_Change      -0.03145392 -0.04441367 -0.01849416
influenza          6.65284195  5.17545509  8.13022881
```

```
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"
```

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	47.015860	0.765811	61.394	<0.001 ***
Triage_Begin_Day == 0	0.012817	0.001194	10.735	<0.001 ***
Level_Change == 0	-4.029599	1.541551	-2.614	0.0382 *
Slope_Change == 0	-0.031454	0.006612	-4.757	<0.001 ***
influenza == 0	6.652842	0.753783	8.826	<0.001 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Day),
method = "REML")
```

Quantile = 2.5101

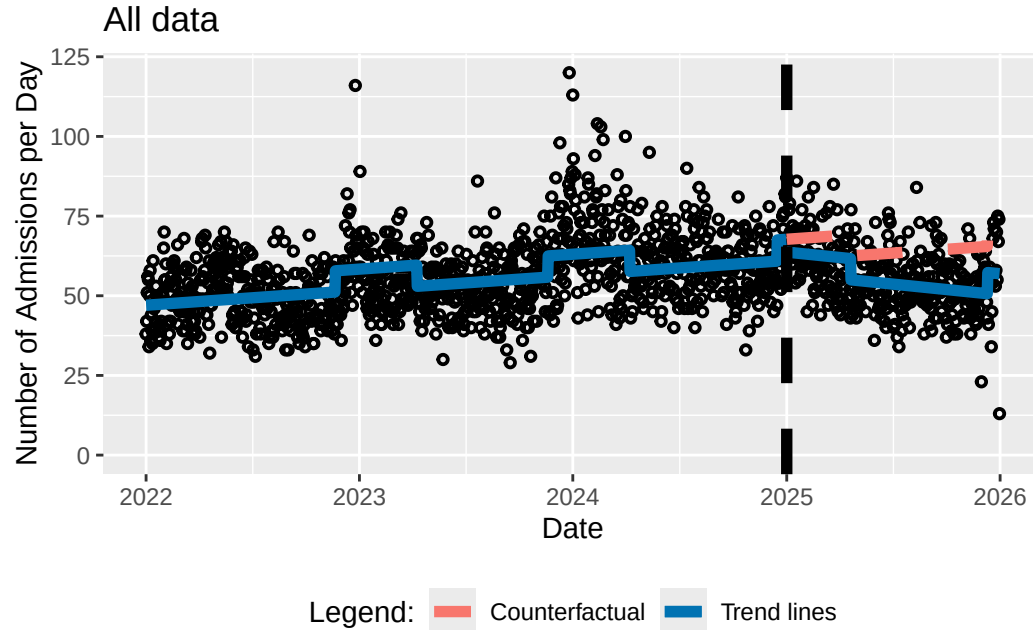
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	47.01586	45.09359	48.93813
Triage_Begin_Day == 0	0.01282	0.00982	0.01581
Level_Change == 0	-4.02960	-7.89906	-0.16014
Slope_Change == 0	-0.03145	-0.04805	-0.01486
influenza == 0	6.65284	4.76076	8.54492

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	47.01586
model_pre_slope	0.01282
model_level_change	-4.02960
model_slope_change	-0.03145
model_influenza	6.65284
model_AIC	10941.15804
model_BIC	10978.14218
model_n	1461.00000



How do I perform the ITSA accounting for influenza using ARMA(1,1) and AR(2) errors?

```
getITSresultsInfARMA11(dat0=datall,outcome="n",ptitle="All data") # ARMA(1,1) - process
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
<environment: 0x000002365e539938>
Generalized least squares fit by REML
```

Model: form
 Data: dat0
 AIC BIC logLik
 10907.1 10949.37 -5445.551

Correlation Structure: ARMA(1,1)
 Formula: ~Triage_Begin_Day
 Parameter estimate(s):
 Phi1 Theta1
 0.9140928 -0.8247787

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	47.57538	1.4829411	32.08177	0.0000
Triage_Begin_Day	0.01115	0.0035436	3.14540	0.0017
Level_Change	7.25721	2.0327438	3.57015	0.0004
Slope_Change	-0.02719	0.0051426	-5.28739	0.0000
influenza	5.13714	1.0949952	4.69148	0.0000

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.337	-0.580		
Slope_Change	0.592	-0.730	-0.004	
influenza	-0.028	-0.180	-0.029	0.226

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-4.2283823	-0.6934942	-0.0554386	0.6210806	5.7754987

Residual standard error: 10.26509

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	47.57538065	44.66886951	50.48189179
Triage_Begin_Day	0.01114612	0.00420074	0.01809150
Level_Change	7.25720730	3.27310263	11.24131197
Slope_Change	-0.02719108	-0.03727045	-0.01711171
influenza	5.13714374	2.99099255	7.28329493

[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"

Simultaneous Tests for General Linear Hypotheses

Fit: gls(model = form, data = dat0, correlation = corARMA(form = ~Triage_Begin_Day,
p = 1, q = 1), method = "REML")

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	47.575381	1.482941	32.082	< 0.001 ***
Triage_Begin_Day == 0	0.011146	0.003544	3.145	0.00767 **
Level_Change == 0	7.257207	2.032744	3.570	0.00146 **
Slope_Change == 0	-0.027191	0.005143	-5.287	< 0.001 ***
influenza == 0	5.137144	1.094995	4.691	< 0.001 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

Fit: gls(model = form, data = dat0, correlation = corARMA(form = ~Triage_Begin_Day,
p = 1, q = 1), method = "REML")

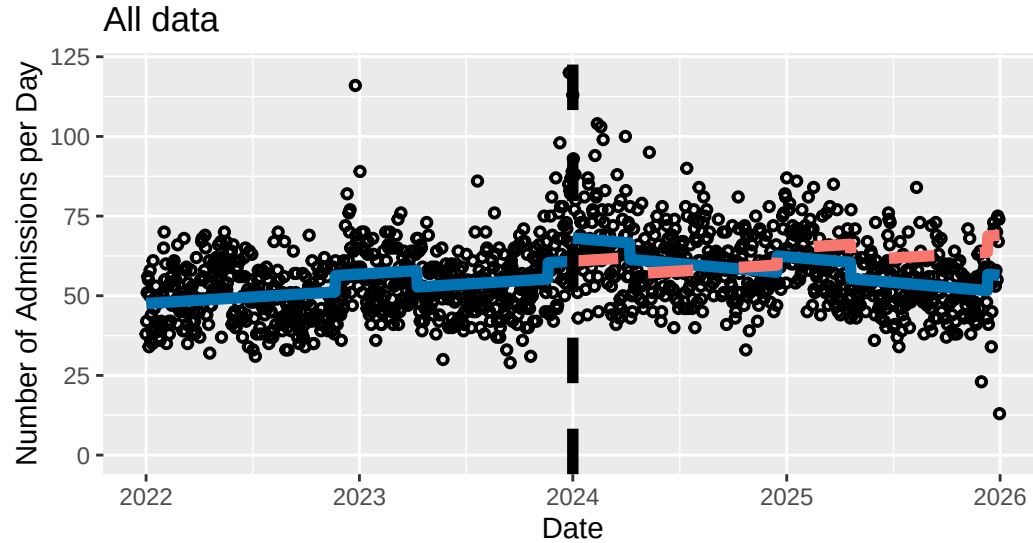
Quantile = 2.5038
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	47.575381	43.862431	51.288331
Triage_Begin_Day == 0	0.011146	0.002274	0.020019
Level_Change == 0	7.257207	2.167675	12.346739
Slope_Change == 0	-0.027191	-0.040067	-0.014315
influenza == 0	5.137144	2.395523	7.878765

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	47.57538
model_pre_slope	0.01115
model_level_change	7.25721
model_slope_change	-0.02719
model_influenza	5.13714
model_AIC	10907.10117
model_BIC	10949.36876
model_n	1461.00000



Legend: Counterfactual Trend lines

```
getITSresultsInfAR2(dat0=datall,outcome="n",ptitle="All data") # ARMA(2,0) - process
```

```
n ~ Triage_Begin_Day + Level_Change + Slope_Change + influenza
```

```
<environment: 0x0000023659ba7628>
```

```
Generalized least squares fit by REML
```

```
Model: form
```

```
Data: dat0
```

```
AIC      BIC      logLik
```

```
10922.89 10965.15 -5453.443
```

```
Correlation Structure: ARMA(2,0)
```

Formula: ~Triage_Begin_Day

Parameter estimate(s):

	Phi1	Phi2
	0.16738381	0.03131965

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	47.81368	0.9301976	51.40164	0
Triage_Begin_Day	0.00979	0.0022440	4.36401	0
Level_Change	8.04053	1.3127294	6.12504	0
Slope_Change	-0.02630	0.0032229	-8.16120	0
influenza	5.81696	0.7477546	7.77924	0

Correlation:

	(Intr)	Tr_B_D	Lvl_Ch	Slp_Ch
Triage_Begin_Day	-0.844			
Level_Change	0.355	-0.593		
Slope_Change	0.584	-0.721	-0.008	
influenza	-0.033	-0.190	-0.042	0.257

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-4.28242134	-0.70197210	-0.05916388	0.63034730	5.79784314

Residual standard error: 10.22266

Degrees of freedom: 1461 total; 1456 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	47.813682027	45.990528193	49.63683586
Triage_Begin_Day	0.009793042	0.005394796	0.01419129
Level_Change	8.040525380	5.467622979	10.61342778

```
Slope_Change      -0.026302811 -0.032619597 -0.01998603
influenza          5.816963236  4.351391141  7.28253533
```

```
[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"
```

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corARMA(form = ~Triage_Begin_Day,
  p = 2, q = 0), method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept) == 0	47.813682	0.930198	51.402	<0.001 ***
Triage_Begin_Day == 0	0.009793	0.002244	4.364	<0.001 ***
Level_Change == 0	8.040525	1.312729	6.125	<0.001 ***
Slope_Change == 0	-0.026303	0.003223	-8.161	<0.001 ***
influenza == 0	5.816963	0.747755	7.779	<0.001 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Adjusted p values reported -- single-step method)
```

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corARMA(form = ~Triage_Begin_Day,
  p = 2, q = 0), method = "REML")
```

```
Quantile = 2.5061
```

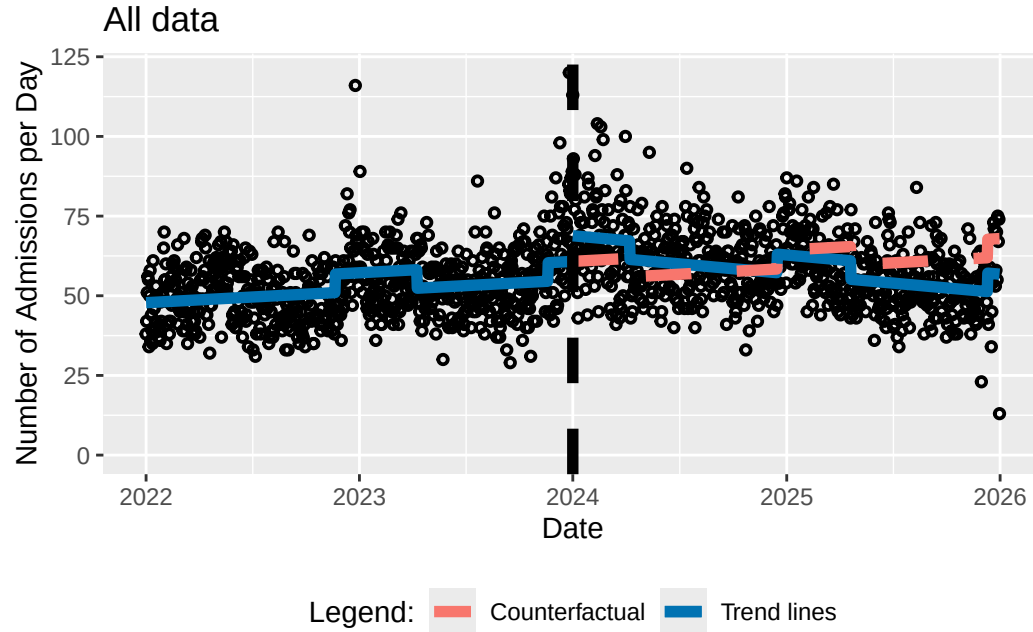
```
95% family-wise confidence level
```

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	47.813682	45.482487	50.144877
Triage_Begin_Day == 0	0.009793	0.004169	0.015417
Level_Change == 0	8.040525	4.750656	11.330395
Slope_Change == 0	-0.026303	-0.034380	-0.018226
influenza == 0	5.816963	3.942994	7.690933

[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	47.81368
model_pre_slope	0.00979
model_level_change	8.04053
model_slope_change	-0.02630
model_influenza	5.81696
model_AIC	10922.88649
model_BIC	10965.15408
model_n	1461.00000



How do I perform the ITSA with Months?

```
getITSresultsMonth(dat0=datallm,outcome="n",ptitle="All data")
```

```
n ~ Triage_Begin_Month + Level_Change + Slope_Change
<environment: 0x000002365e3b0390>
Generalized least squares fit by REML
Model: form
Data: dat0
```

AIC	BIC	logLik
583.2494	593.9545	-285.6247

Correlation Structure: AR(1)
 Formula: ~Triage_Begin_Month
 Parameter estimate(s):
 Phi
 0.3817266

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	1422.3023	83.64773	17.003478	0.0000
Triage_Begin_Month	15.5313	5.73413	2.708562	0.0096
Level_Change	148.7655	105.62948	1.408371	0.1660
Slope_Change	-30.2976	8.47780	-3.573754	0.0009

Correlation:

	(Intr)	Tr_B_M	Lvl_Ch
Triage_Begin_Month	-0.865		
Level_Change	0.338	-0.591	
Slope_Change	0.619	-0.739	0.040

Standardized residuals:

Min	Q1	Med	Q3	Max
-1.5014169	-0.5776937	-0.2152613	0.5403930	2.6630722

Residual standard error: 141.9217

Degrees of freedom: 48 total; 44 residual

[1] "Below are the model estimates together with their 95% confidence limits."

	Point estimates	2.5 %	97.5 %
(Intercept)	1422.30228	1258.35575	1586.24882

Triage_Begin_Month	15.53125	4.29256	26.76995
Level_Change	148.76551	-58.26447	355.79550
Slope_Change	-30.29756	-46.91374	-13.68138

[1] "Below are the model estimates together with multiplicity adjusted p-values and confidence limits"

Simultaneous Tests for General Linear Hypotheses

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Month),
method = "REML")
```

Linear Hypotheses:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept) == 0	1422.302	83.648	17.003	<0.001	***
Triage_Begin_Month == 0	15.531	5.734	2.709	0.0204	*
Level_Change == 0	148.766	105.629	1.408	0.3810	
Slope_Change == 0	-30.298	8.478	-3.574	<0.001	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

Simultaneous Confidence Intervals

```
Fit: gls(model = form, data = dat0, correlation = corAR1(form = ~Triage_Begin_Month),
method = "REML")
```

Quantile = 2.3963

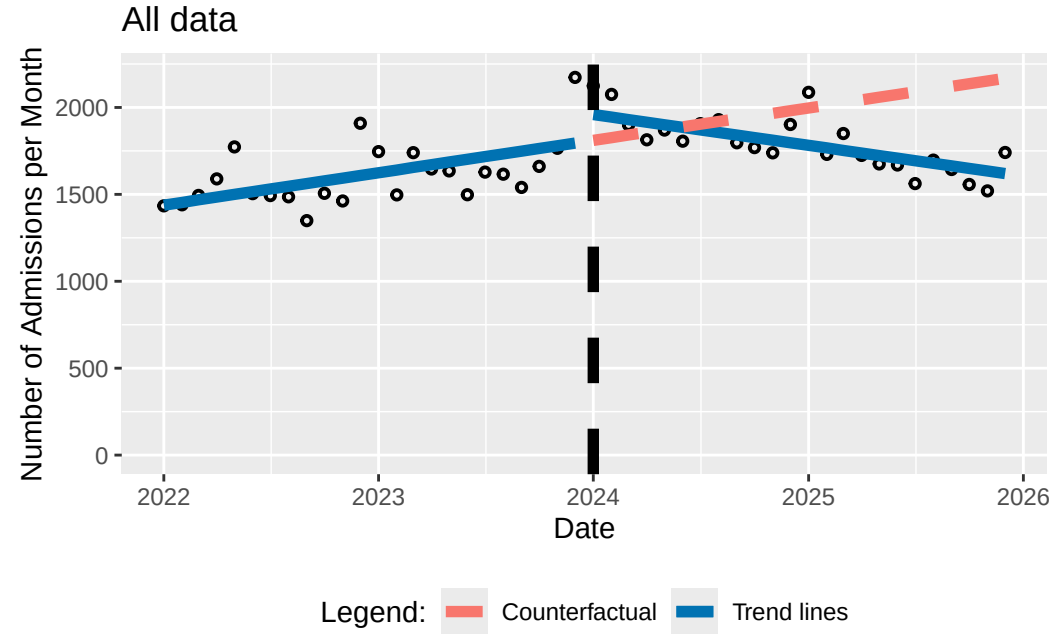
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
(Intercept) == 0	1422.3023	1221.8577	1622.7469
Triage_Begin_Month == 0	15.5313	1.7906	29.2719
Level_Change == 0	148.7655	-104.3538	401.8849
Slope_Change == 0	-30.2976	-50.6129	-9.9823

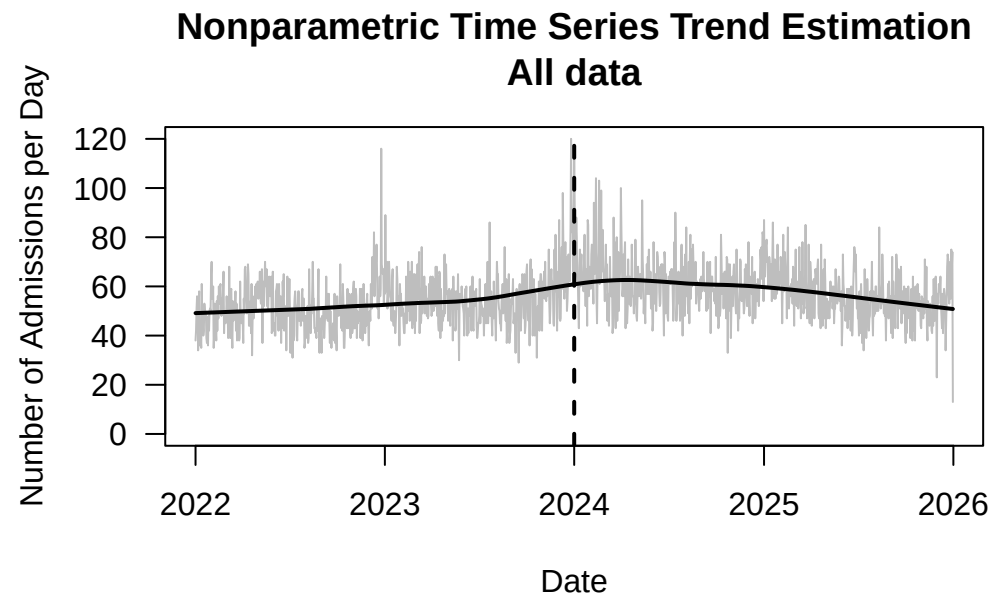
[1] "Below is the output requested by Alexander Braun"

	(Intercept)
model_intercept	1422.30228
model_pre_slope	15.53125
model_level_change	148.76551
model_slope_change	-30.29756
model_AIC	583.24937
model_BIC	593.95451
model_n	48.00000

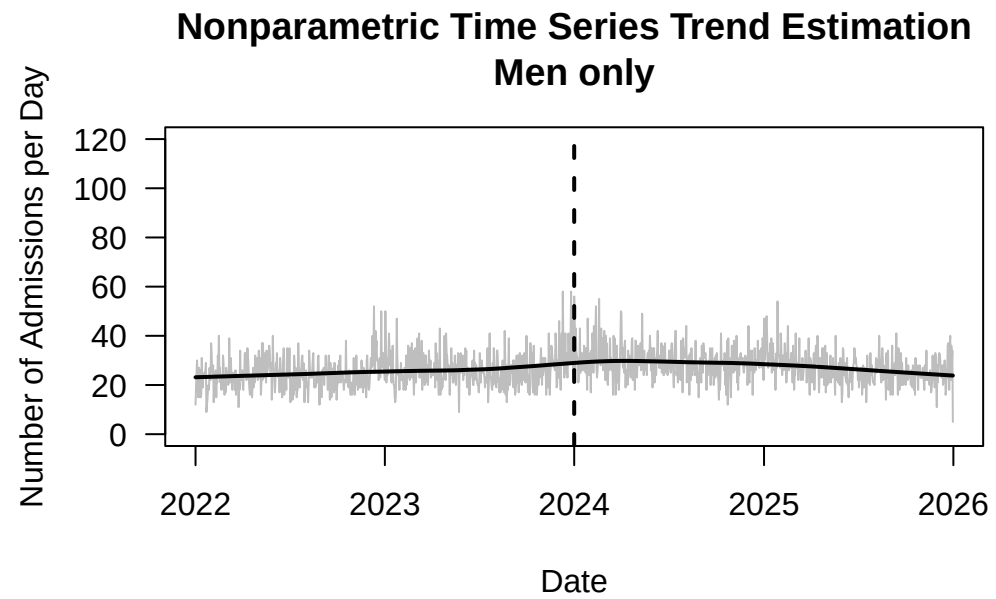


How do I estimate the time series trend using the semiparametric model suggested by Feng et al.?

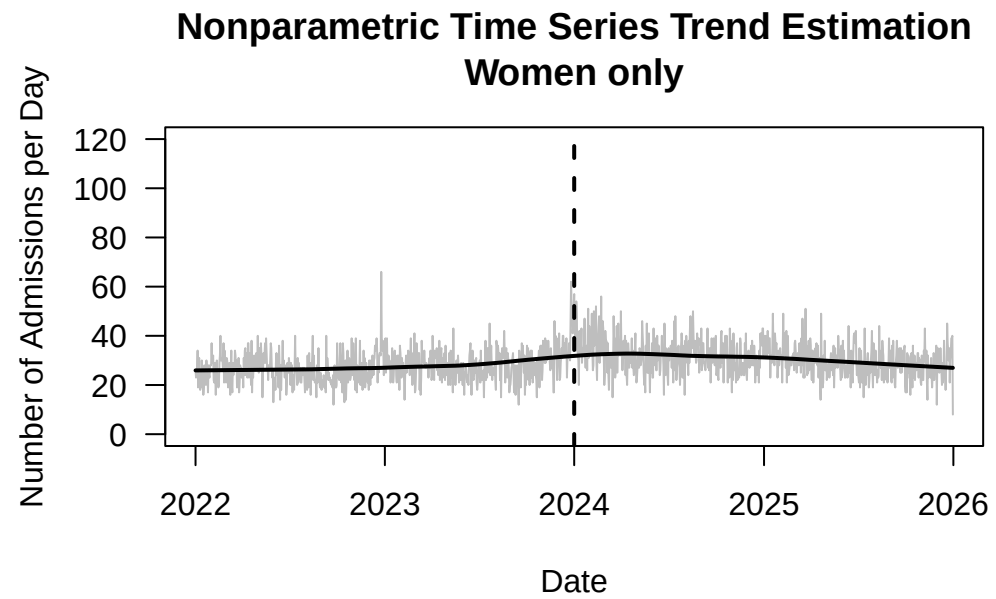
```
getnts(dat0=datall,ptitle="All data")
```



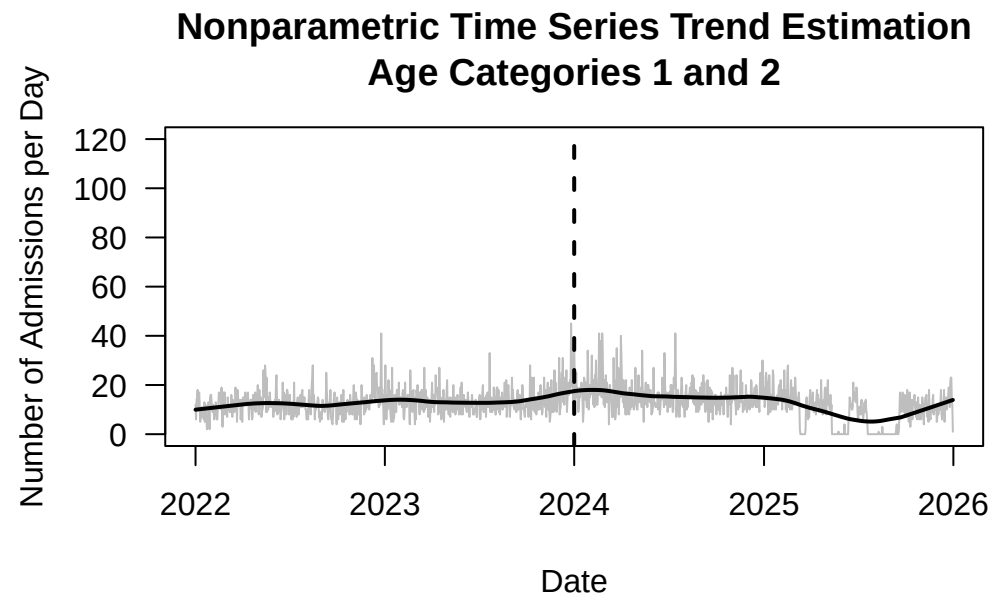
```
getnts(dat0=datsem,ptitle="Men only")
```



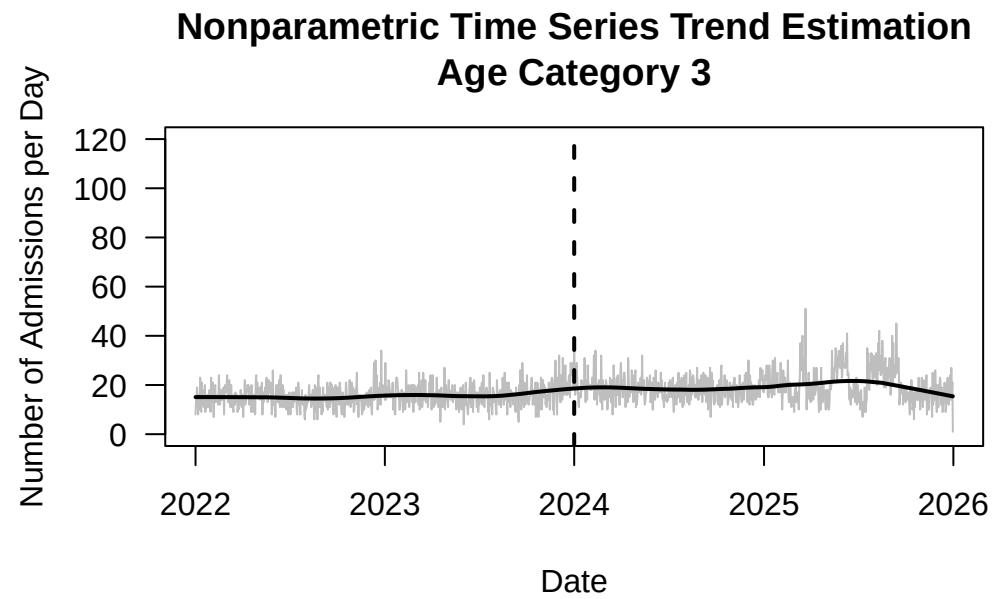
```
getnts(dat0=datsef,ptitle="Women only")
```



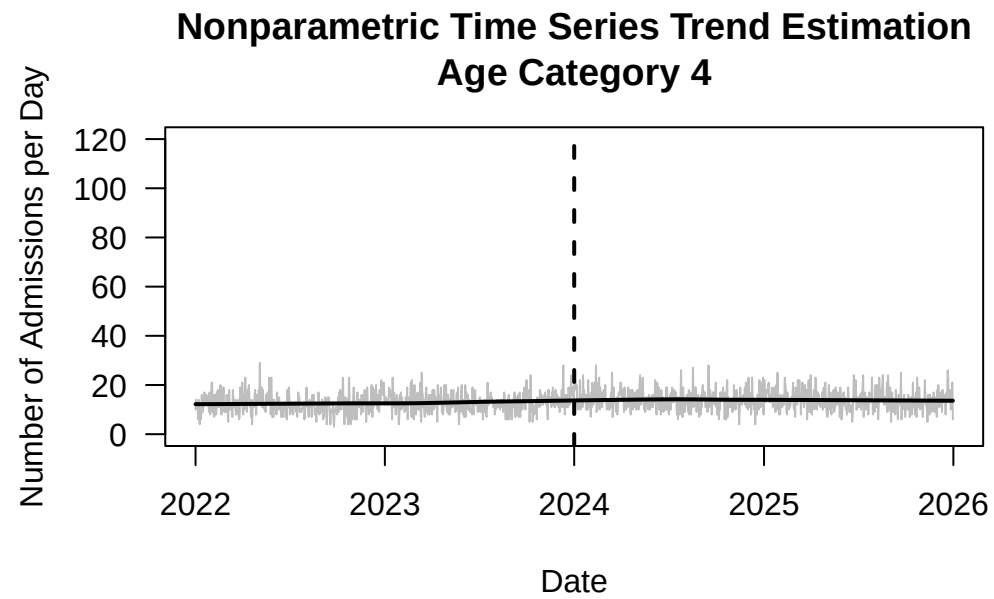
```
getnts(dat0=datag12,ptitle="Age Categories 1 and 2")
```

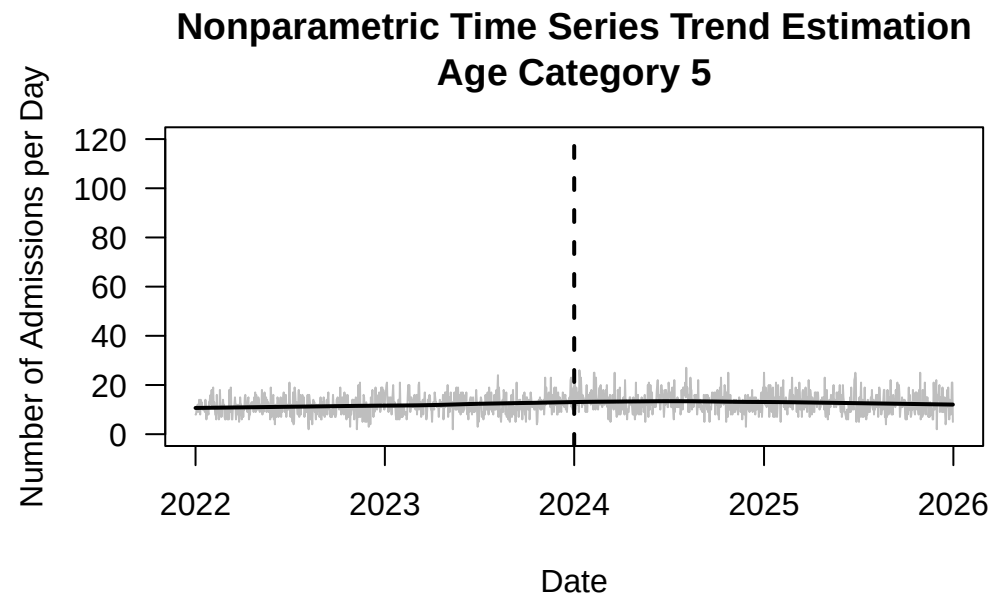
```
getnts(dat0=datag3,ptitle="Age Category 3")
```



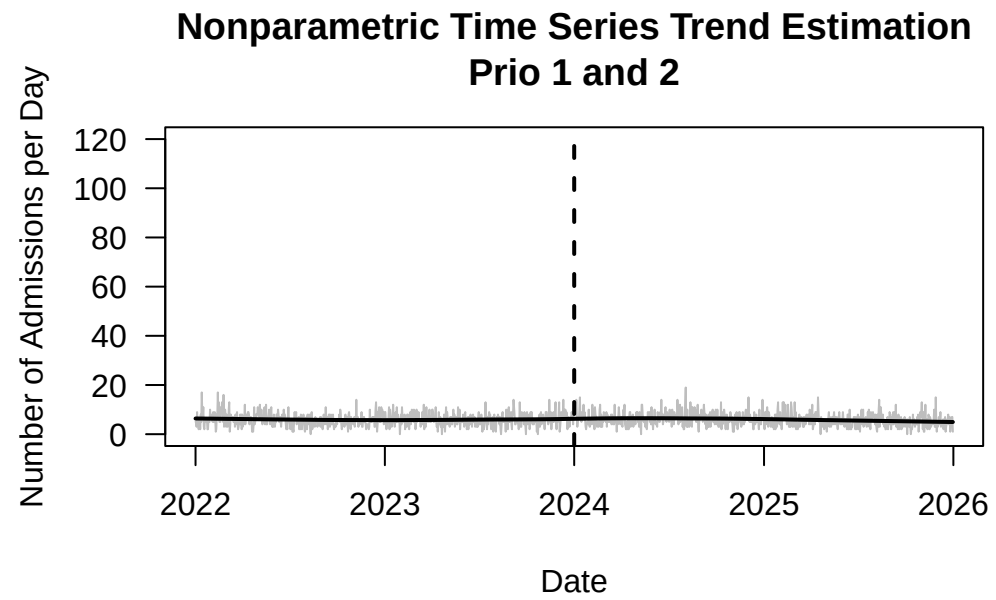
```
getnts(dat0=datag4,ptitle="Age Category 4")
```



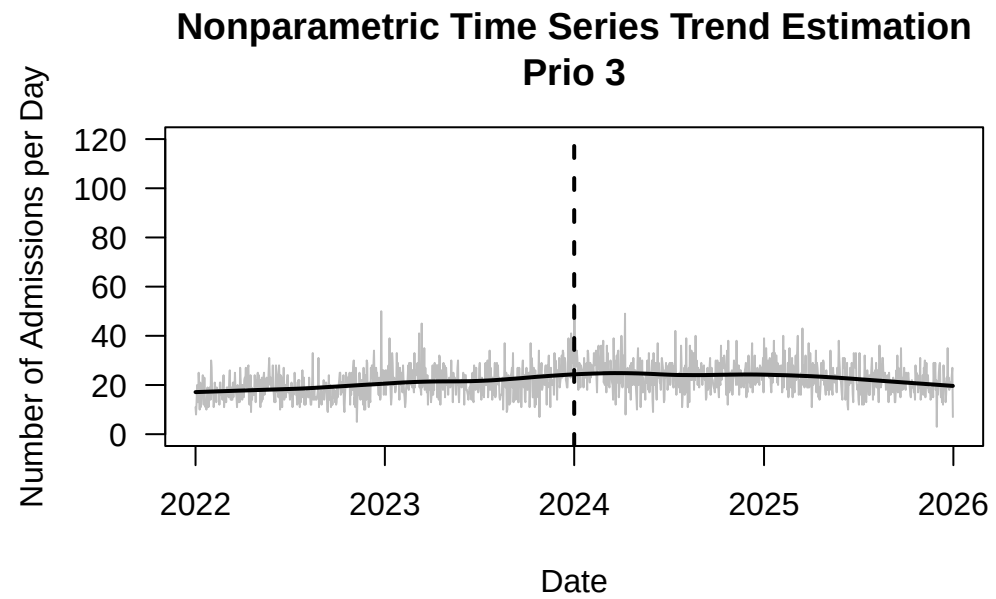
```
getnts(dat0=datag5,ptitle="Age Category 5")
```



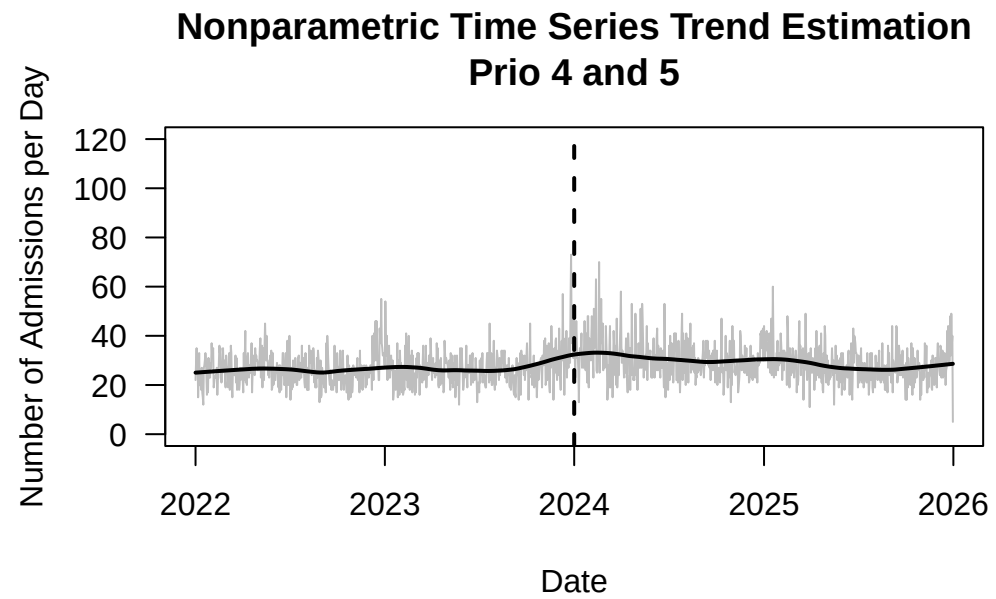
```
getnts(dat0=datpr12,ptitle="Prio 1 and 2")
```



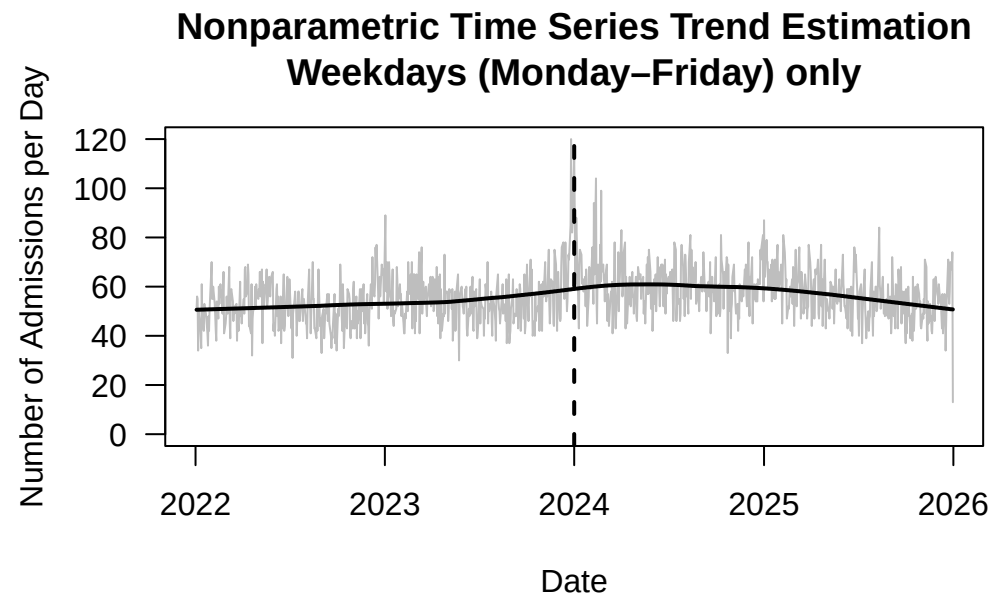
```
getnts(dat0=datpr3,ptitle="Prio 3")
```



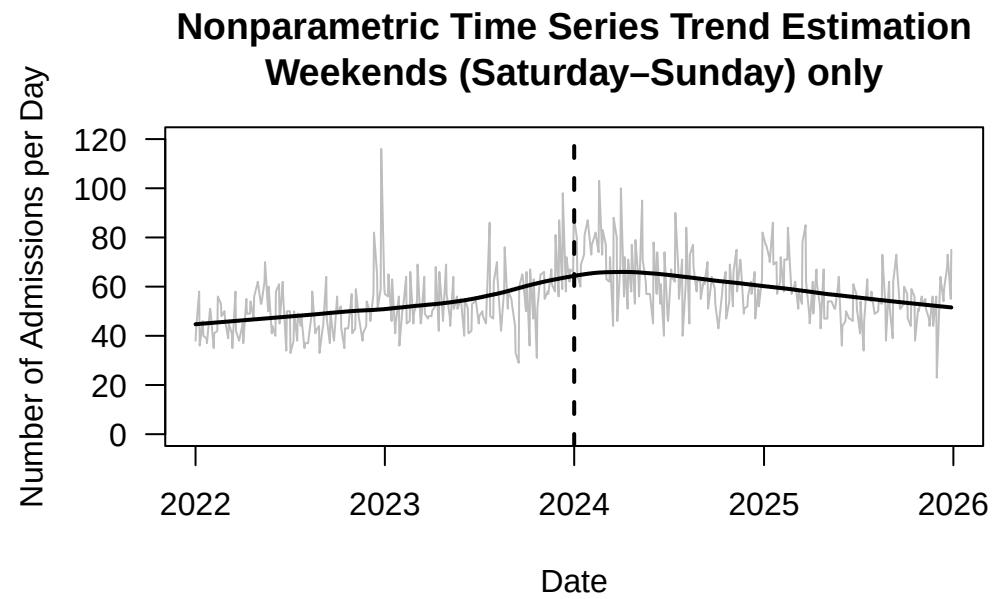
```
getnts(dat0=datpr45,ptitle="Prio 4 and 5")
```



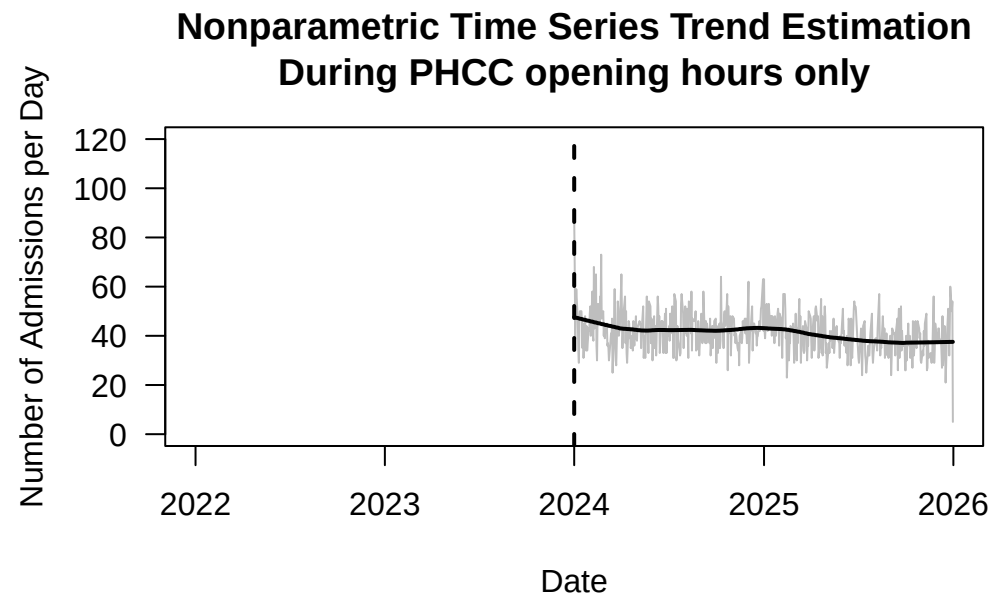
```
getnts(dat0=datwk,ptitle="Weekdays (Monday-Friday) only")
```



```
getnts(dat0=datss,ptitle="Weekends (Saturday–Sunday) only")
```

```
getnts(dat0=datop,ptitle="During PHCC opening hours only")
```



```
getnts(dat0=datopn,ptitle="Outside PHCC opening hours only")
```

